

Why Mathematics Works: Structural Necessity of Isomorphism Between Formal Systems and Physical Reality

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Abstract

Eugene Wigner’s celebrated 1960 essay posed a question that remains open: why is mathematics so unreasonably effective in the natural sciences? This paper proposes a resolution grounded in the theory of persistent systems. The central argument is that the effectiveness of mathematics is not a mysterious pre-established harmony between mind and cosmos, but a *structural necessity* arising from the conditions that any system must satisfy in order to persist through time. We show that persistence requires the existence of conserved quantities and invariant relations, and that any collection of such invariants necessarily constitutes a rich algebraic structure—finitely axiomatizable, possessed of non-trivial consequences, and multiply realizable across diverse physical instantiations. This algebraic structure is, by construction, a model of a substantive formal theory, thereby guaranteeing a systematic correspondence between the physical system and a mathematical description. The argument is developed through a sequence of formally stated definitions, propositions, and theorems, each accompanied by extended interpretive discussion accessible to researchers across disciplines, and illustrated by five worked models spanning discrete, chaotic, cellular-automaton, dissipative, and biological systems. We distinguish this *persistence filter thesis* from Tegmark’s Mathematical Universe Hypothesis, structural realism, the mapping account, conventionalism, and information-theoretic approaches. We confront the Kantian counter-thesis—that mathematics is a coordinate grid imposed by consciousness rather than a reflection of external structure—proving it consistent but insufficient, and arriving at a dual-source decomposition of mathematical physics into cognitive and

empirical components. The paper addresses nine counterarguments, identifies six falsifiable predictions, and delineates the scope, limitations, and open problems of the framework.

Keywords: philosophy of mathematics, unreasonable effectiveness, Wigner's puzzle, persistent systems, structural realism, symmetry, conservation laws, category theory, mathematical physics, invariant algebra, falsifiable predictions

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1 Introduction

In 1960, the physicist and Nobel laureate Eugene Wigner published an essay that has haunted the philosophy of science ever since [Wigner, 1960]. Entitled “The Unreasonable Effectiveness of Mathematics in the Natural Sciences,” the essay posed a deceptively simple question: why do the abstract structures invented by mathematicians—often for purely aesthetic or logical reasons, with no physical application in mind—turn out to describe the physical world with extraordinary precision?

The question is not merely rhetorical. Complex numbers, developed centuries before quantum mechanics, turned out to be indispensable for describing quantum states. Riemannian geometry, created by Bernhard Riemann in the 1850s as a purely mathematical exercise, became the language of Einstein’s general relativity sixty years later. Group theory, initially the study of symmetries of algebraic equations, now provides the foundational framework for the Standard Model of particle physics. These are not isolated coincidences; they constitute a systematic pattern that demands explanation.

The existing responses to Wigner’s puzzle fall into several broad categories, none of which has achieved consensus. *Mathematical Platonism* holds that the physical world *is* a mathematical structure, so the effectiveness of mathematics is simply a tautology—we are describing the world in its own language [Tegmark, 2008]. *Naturalism* and *evolutionary epistemology* suggest that our mathematical intuitions have been shaped by natural selection to track physical regularities [Dennett, 1995]. *Selection bias* arguments propose that we remember the mathematical theories that work and forget the vast majority that do not [Hamming, 1980]. *Conventionalism* argues that the “fit” between mathematics and physics is partly an artifact of how we construct our physical theories [Poincaré, 1902].

Each of these responses captures something important, yet each leaves significant residual puzzlement. Platonism explains effectiveness at the cost of ontological extravagance. Evolutionary accounts explain rough quantitative reasoning but not the precision of fundamental physics. Selection bias cannot account for the *depth* of mathematical applicability—the way a single mathematical structure illuminates vast domains of physical phenomena simultaneously. Conventionalism struggles with cases where mathematical structure *predicts* genuinely novel physical phenomena, as when the Dirac equation predicted antimatter.

This paper proposes a different kind of answer. Rather than asking what special property of mathematics or of the physical world creates the correspondence between them, we ask: *what structural features must any system possess in order to exist stably through time?* We argue that the answer to this question—formalized through the theory of persistent systems—simultaneously explains why such systems are necessarily amenable to mathematical description.

The core argument proceeds as follows:

- (i) Any physical system that persists through time must possess quantities or relations that remain invariant under the transformations the system undergoes.
- (ii) Any collection of invariants under a family of transformations necessarily carries algebraic structure—minimally, the structure of a group action, and typically much richer structure.

- (iii) Any algebraic structure is a model of some first-order theory (this observation, taken alone, is near-trivial—see Section 4.3 for the essential refinement). The non-trivial content of our argument is that the algebraic structure arising from persistence is *rich*: it is finitely generated, satisfies non-trivial polynomial identities, and encodes the symmetry group of the dynamics. It is therefore a model of a *substantive* mathematical theory—one with genuine predictive and classificatory power—not merely of the vacuous theory that lists the structure’s own diagram.
- (iv) Therefore, every persistent physical system admits a faithful mathematical description of a kind that supports prediction, unification, and structural classification—the hallmarks of the mathematical physics that prompted Wigner’s puzzle.

We call this the **Persistence Filter Thesis**: mathematics works not because the universe is intrinsically “mathematical” in some Platonic sense, but because only those configurations of matter and energy that possess the right kind of structural regularity—the kind that admits mathematical description—survive the filter of temporal persistence.

The paper is organized as follows. Section 2 reviews the necessary background from dynamical systems theory, abstract algebra, and mathematical logic, presenting each concept with extended discussion to ensure accessibility across disciplines. Section 3 develops the formal theory of persistent systems and their invariant structure. Section 4 contains the main results: the formal derivation of the Persistence Filter Thesis and the proof that the isomorphism between persistent systems and formal theories is structurally necessary. Section 5 addresses the specificity problem—why *particular* mathematical structures recur in physics. Section 6 provides five illustrative models—discrete, chaotic, cellular-automaton, dissipative, and biological—that make the abstract argument concrete. Section 7 situates the thesis within the broader literature on mathematical applicability, engaging with Platonist, structuralist, naturalistic, mapping-theoretic, conventionalist, information-theoretic, and category-theoretic approaches. Section 8 discusses implications for biology, the social sciences, and the foundations of mathematics. Section 9 confronts a radical alternative—the possibility that mathematics is not a reflection of the world but a coordinate grid imposed by consciousness on its own internal model—developing this counter-thesis formally, refuting it with equal rigor, and arriving at a synthesis that decomposes mathematical physics into cognitive and empirical components. Section 10 addresses nine counterarguments, including the charges of vacuity, circularity, and unfalsifiability. Section 11 identifies six falsifiable predictions of the thesis. Section 12 delineates the scope and limitations of the argument, and Section 14 concludes with open problems.

2 Background and Preliminaries

This section establishes the conceptual and mathematical vocabulary required for the main argument. Each concept is introduced with motivation and interpretive commentary, so that readers from diverse disciplinary backgrounds can follow the formal development in subsequent sections.

2.1 Dynamical Systems: The Language of Change

The physical world is characterized by change. Planets orbit, populations fluctuate, chemical reactions proceed, economies cycle. The mathematical framework that captures the general structure of change is the theory of *dynamical systems*.

Definition 2.1 (Dynamical System). A **dynamical system** is a triple (X, T, Φ) , where:

- X is a set called the *state space*, whose elements represent all possible states of the system;
- T is a monoid (typically $\mathbb{R}$, $\mathbb{R}_{\geq 0}$, $\mathbb{Z}$, or $\mathbb{N}$), called the *time set*, representing the instants at which the system is observed;
- $\Phi : T \times X \rightarrow X$ is the *evolution operator*, satisfying:

$$\Phi(0, x) = x \quad \text{for all } x \in X, \quad (1)$$

$$\Phi(t, \Phi(s, x)) = \Phi(t + s, x) \quad \text{for all } s, t \in T, x \in X. \quad (2)$$

To unpack this definition: the state space X is the collection of all configurations the system can occupy. For a pendulum, X might consist of all possible combinations of angle and angular velocity. For an ecological community, X might describe the population sizes of all species in the community. The evolution operator Φ tells us how the system changes: given that the system is in state x at some reference time, $\Phi(t, x)$ is the state it will occupy after an additional time interval t has elapsed.

Equation (1) says that zero elapsed time produces no change—the system stays where it is. Equation (2) says that evolving for time s and then for time t produces the same result as evolving for time $s + t$ in a single step. This is the *semigroup property*, and it captures the idea that the laws governing the system do not themselves change over time—the rule for evolving from any state is always the same.

Example 2.2 (Newtonian Particle). Consider a particle of mass m moving in one dimension under a force $F(x)$. The state space is $X = \mathbb{R}^2$, with coordinates (x, v) representing position and velocity. The evolution is governed by Newton’s second law:

$$\frac{dx}{dt} = v, \quad \frac{dv}{dt} = \frac{F(x)}{m}. \quad (3)$$

The evolution operator $\Phi(t, (x_0, v_0))$ maps the initial condition (x_0, v_0) to the state $(x(t), v(t))$ obtained by solving these equations forward in time by t units.

Example 2.3 (Population Dynamics). In the Lotka–Volterra model of predator–prey interaction, the state space is $X = \mathbb{R}_{\geq 0}^2$, representing the population sizes (N, P) of prey and predator respectively. The evolution follows:

$$\frac{dN}{dt} = \alpha N - \beta NP, \quad \frac{dP}{dt} = \delta NP - \gamma P, \quad (4)$$

where $\alpha, \beta, \gamma, \delta > 0$ are parameters encoding birth rates, predation rates, and death rates. Here the state space is continuous, the dynamics are nonlinear, and the system exhibits characteristic oscillatory behavior.

2.2 Symmetry and Group Actions

One of the most profound discoveries of modern science is that the concept of *symmetry*—the idea that something remains unchanged under a transformation—lies at the heart of physical law. This insight, developed over the course of the nineteenth and twentieth centuries, connects the everyday notion of symmetry (a snowflake looks the same after rotation by 60°) to the deepest structures of fundamental physics.

Definition 2.4 (Group). A **group** is a set G equipped with a binary operation $\cdot : G \times G \rightarrow G$ satisfying three axioms:

- (a) **Associativity:** $(g \cdot h) \cdot k = g \cdot (h \cdot k)$ for all $g, h, k \in G$.
- (b) **Identity:** There exists an element $e \in G$ such that $e \cdot g = g \cdot e = g$ for all $g \in G$.
- (c) **Inverses:** For every $g \in G$, there exists $g^{-1} \in G$ such that $g \cdot g^{-1} = g^{-1} \cdot g = e$.

The concept of a group captures the essence of reversible transformation. Condition (a) says that the order of composition does not matter when combining three transformations. Condition (b) says there is a “do nothing” transformation. Condition (c) says every transformation can be undone. Together, these axioms encode the minimal structure that any collection of reversible operations must possess.

Definition 2.5 (Group Action). A **(left) group action** of a group G on a set X is a map $\alpha : G \times X \rightarrow X$, often written $\alpha(g, x) = g \cdot x$, satisfying:

$$e \cdot x = x \quad \text{for all } x \in X, \tag{5}$$

$$g \cdot (h \cdot x) = (g \cdot h) \cdot x \quad \text{for all } g, h \in G, x \in X. \tag{6}$$

A group action formalizes the idea that the transformations in G can be “applied to” the elements of X . Equation (5) says the identity transformation leaves everything unchanged. Equation (6) says that applying h and then g is the same as applying the composite transformation $g \cdot h$.

Definition 2.6 (Invariant). Given a group G acting on a set X , a function $f : X \rightarrow Y$ (where Y is any set) is a **G -invariant** if:

$$f(g \cdot x) = f(x) \quad \text{for all } g \in G, x \in X. \tag{7}$$

The collection of all G -invariant functions on X (with values in a fixed ring R) is denoted $\text{Inv}_G(X, R)$.

An invariant is a quantity that does not change when the system is transformed. The total energy of an isolated system does not change over time. The total electric charge in a closed region does not change as particles interact. The speed of light does not change when one switches between inertial reference frames. Each of these is an invariant with respect to a specific group of transformations, and each encodes a fundamental regularity of the physical world.

2.3 Conservation Laws and Noether's Theorem

The connection between symmetry and conservation is one of the deepest results in mathematical physics, established by Emmy Noether in 1918 [Noether, 1918].

Theorem 2.7 (Noether's Theorem — Informal Statement). *Every continuous symmetry of the action of a physical system corresponds to a conserved quantity. Conversely, every conserved quantity corresponds to a continuous symmetry.*

This theorem establishes a precise correspondence: time-translation symmetry (the laws of physics are the same today as yesterday) corresponds to conservation of energy; spatial-translation symmetry (the laws are the same here as there) corresponds to conservation of momentum; rotational symmetry corresponds to conservation of angular momentum.

For the formal statement, consider a Lagrangian system with generalized coordinates $q = (q_1, \dots, q_n)$ and Lagrangian $L(q, \dot{q}, t)$. The action functional is:

$$S[q] = \int_{t_1}^{t_2} L(q(t), \dot{q}(t), t) dt. \quad (8)$$

Theorem 2.8 (Noether's Theorem — Formal Statement). *Let q^s denote a one-parameter family of transformations of the generalized coordinates, with $q^0 = q$ the identity. If the action S is invariant under this family, that is, if $\frac{d}{ds}\big|_{s=0} L(q^s, \dot{q}^s, t) = \frac{d}{dt} \Lambda$ for some function Λ , then the quantity*

$$Q = \sum_{i=1}^n \frac{\partial L}{\partial \dot{q}_i} \frac{\partial q_i^s}{\partial s} \bigg|_{s=0} - \Lambda \quad (9)$$

is conserved along solutions of the Euler–Lagrange equations, that is, $\frac{dQ}{dt} = 0$.

The quantity Q defined in Equation (9) is called the *Noether charge* associated with the symmetry. The summation involves the *generalized momenta* $\frac{\partial L}{\partial \dot{q}_i}$ (these are the momenta conjugate to each coordinate) and the *infinitesimal generators* $\frac{\partial q_i^s}{\partial s}\big|_{s=0}$ (these describe how each coordinate changes under an infinitesimally small symmetry transformation).

The profound content of Noether's theorem, for our purposes, is this: for continuous symmetries of Lagrangian systems, symmetry and conservation are not merely correlated—they are *mathematically equivalent*. Wherever there is a continuous symmetry, there is a conserved quantity, and wherever there is a conserved quantity, there is a continuous symmetry.

Two important qualifications are necessary for the generality of our later argument. First, Noether's theorem as stated applies to *continuous* symmetries of systems admitting a *Lagrangian formulation*. Discrete symmetries (such as parity, charge conjugation, and time reversal) do not produce conservation laws via the Noether mechanism; they produce *selection rules* and *quantum numbers* instead [Weinberg, 1995]. Second, dissipative systems—those with friction, viscosity, or other irreversible processes—do not conserve energy or other classical Noether charges. For such systems, the invariants that underwrite persistence are not strict conservation laws but rather attractors, Lyapunov functions, and dissipative balance relations (such as the entropy production relations of non-equilibrium thermodynamics) [Strogatz, 2015].

The persistence filter thesis, as developed in Sections 3 and 4, does *not* depend on Noether’s theorem specifically. It requires only that persistent systems possess invariant quantities—not that those invariants arise through the Noether mechanism. Noether’s theorem is invoked here as the most celebrated and elegant illustration of the symmetry-invariance correspondence, and as the source of exact conservation laws in fundamental physics. But the algebraic argument of Section 3—that invariants form an algebra, which encodes group structure—holds regardless of whether the invariants are Noether charges, discrete quantum numbers, topological invariants, or Lyapunov functions. The generality of the persistence filter exceeds the scope of any single theorem connecting symmetry to conservation.

2.4 Algebraic Structures and Their Models

We now review the key concept from mathematical logic that bridges the gap between abstract mathematical structures and the physical systems that realize them.

Definition 2.9 (First-Order Language and Theory). A **first-order language** $\mathcal{L}$ consists of:

- a set of *constant symbols* $c_1, c_2, \dots$;
- a set of *function symbols* $f_1, f_2, \dots$, each with a specified arity (number of arguments);
- a set of *relation symbols* $R_1, R_2, \dots$, each with a specified arity;
- the usual logical connectives ($\wedge, \vee, \neg, \rightarrow, \leftrightarrow$), quantifiers ($\forall, \exists$), variables, and parentheses.

A **first-order theory** $\mathcal{T}$ in the language $\mathcal{L}$ is a set of sentences (formulas with no free variables) in $\mathcal{L}$, called the *axioms* of $\mathcal{T}$.

This definition describes the formal machinery of mathematical theories in their most general form. A first-order language provides the vocabulary—the nouns (constants), verbs (functions), and adjectives (relations)—while a theory provides the grammar and content—the axioms that constrain how these symbols can relate to one another.

Definition 2.10 (Model). A **model** (or **structure**) $\mathcal{M}$ for a first-order language $\mathcal{L}$ consists of:

- a non-empty set M called the *domain* (or *universe*) of $\mathcal{M}$;
- for each constant symbol c in $\mathcal{L}$, an element $c^{\mathcal{M}} \in M$;
- for each n -ary function symbol f in $\mathcal{L}$, a function $f^{\mathcal{M}} : M^n \rightarrow M$;
- for each n -ary relation symbol R in $\mathcal{L}$, a subset $R^{\mathcal{M}} \subseteq M^n$.

A model $\mathcal{M}$ is a **model of a theory** $\mathcal{T}$ if every axiom of $\mathcal{T}$ is true in $\mathcal{M}$ (under the standard interpretation of logical connectives and quantifiers).

The concept of a model is the formal counterpart of the intuitive idea that an abstract mathematical structure can be “realized” or “instantiated” by concrete objects. The integers with addition form a model of the theory of abelian groups. The rotations of three-dimensional space form a model of the theory of $\text{SO}(3)$. A population of organisms subject to natural selection can be viewed as a model of certain dynamical systems theories.

Theorem 2.11 (Gödel’s Completeness Theorem). *A first-order theory $\mathcal{T}$ has a model if and only if $\mathcal{T}$ is consistent (that is, no contradiction can be derived from its axioms).*

Gödel’s completeness theorem, proved in 1929 [Gödel, 1930], establishes a fundamental bridge between syntax and semantics: any consistent set of axioms describes *something*—there is always a concrete structure that satisfies those axioms. For our argument, the direction we will use is the converse: any concrete structure with definite properties *is* a model of some consistent theory. The physical world, insofar as it possesses definite, stable properties, is therefore guaranteed to be describable by some formal mathematical theory.

3 The Theory of Persistent Systems

We now develop the formal apparatus of persistent systems. The central idea is straightforward: among all conceivable configurations of matter and energy, only some endure through time. We formalize what it means to endure, and then investigate what structural properties a system must possess in order to do so.

3.1 Persistence: Formal Definition

Definition 3.1 (State Space and Observable). Let X be a state space (a topological space or measurable space, depending on context). An **observable** is a measurable function $\phi : X \rightarrow V$, where V is a topological vector space (typically $\mathbb{R}^n$ for some n). The collection of all observables on X with values in V is denoted $\mathcal{O}(X, V)$.

Observables are the quantities we can measure about a system: its temperature, its position, the size of a population, the concentration of a chemical species, the price of an asset. Each observable assigns a numerical value (or a vector of values) to each possible state of the system.

Definition 3.2 (Persistent System). Let (X, T, Φ) be a dynamical system with state space X , time set T , and evolution operator Φ . Let $\mathcal{O}(X, V)$ denote the space of observables. The system is **(ε, τ) -persistent with respect to a collection of observables $\mathcal{F} \subseteq \mathcal{O}(X, V)$** if there exists a non-empty subset $A \subseteq X$ (the *persistent region*) such that for every initial state $x_0 \in A$, every observable $\phi \in \mathcal{F}$, and every time $t \in [0, \tau]$:

$$\|\phi(\Phi(t, x_0)) - \phi(x_0)\| < \varepsilon. \quad (10)$$

The system is **persistent** if it is (ε, τ) -persistent for some $\varepsilon > 0$ and some τ that is large relative to the characteristic timescales of the system’s dynamics.

This definition captures a simple but important idea: a persistent system is one whose key observable properties remain approximately stable over extended periods. A hydrogen atom persists because its energy levels, charge distribution, and spectral properties remain constant (until it is disturbed). An organism persists because it maintains homeostasis—its temperature, pH, metabolic rates, and structural integrity remain within narrow bounds. A social institution persists because its defining rules, norms, and patterns of behavior remain recognizable over time.

Note that persistence does not require that the system be static. A beating heart is highly dynamic, yet it persists: the observable “cardiac output” fluctuates within a bounded range. What persistence requires is not stasis but *regularity*—the existence of quantities that are maintained, whether exactly or approximately, through the system’s evolution.

Definition 3.3 (Exact Persistence and Conserved Observables). A system is **exactly persistent** with respect to an observable ϕ if:

$$\phi(\Phi(t, x)) = \phi(x) \quad \text{for all } t \in T, x \in A. \quad (11)$$

Such an observable ϕ is called a **conserved observable** (or *first integral*) of the system on A .

Exact persistence is the idealized case—the observable is *exactly* unchanged by the dynamics. The total energy of an isolated Hamiltonian system, the total charge in a closed region, the baryon number of an isolated system—these are all exactly conserved observables. Much of fundamental physics is organized around such exact conservation laws, and our argument will show that this is not a coincidence.

3.2 The Persistence Requires Invariants Lemma

We now establish the first key step in our argument: persistent systems must possess invariants.

Lemma 3.4 (Persistence Requires Invariants). *Let (X, T, Φ) be a dynamical system, and let $A \subseteq X$ be a persistent region. Define the one-parameter family of transformations $\{\Phi_t\}_{t \in T}$ by $\Phi_t(x) := \Phi(t, x)$. If ϕ is a conserved observable on A , then ϕ is an invariant of the family $\{\Phi_t\}$, that is:*

$$\phi \circ \Phi_t = \phi \quad \text{on } A, \text{ for all } t \in T. \quad (12)$$

Conversely, if the system has no invariant observables on A , then it is not exactly persistent with respect to any observable on A .

Proof. The forward direction follows immediately from Definition 3.3: if ϕ is conserved, then $\phi(\Phi_t(x)) = \phi(x)$ for all t and $x \in A$, which is precisely the statement that ϕ is invariant under $\{\Phi_t\}$.

For the converse, suppose the system has no invariant observables. Then for every observable $\phi \in \mathcal{O}(X, V)$, there exists some $t_\phi \in T$ and some $x_\phi \in A$ such that $\phi(\Phi_{t_\phi}(x_\phi)) \neq \phi(x_\phi)$. This means no observable is conserved, so the system is not exactly persistent with respect to any observable. $\square$

The content of this lemma, stripped of technicalities, is this: *to persist is to have something that does not change*. If every measurable property of a system changes over time, then the system does not persist in any meaningful sense—it has no enduring identity. Conversely, whatever it is that makes a system recognizably “the same system” over time must be some quantity or relation that remains invariant.

This observation, though simple, is the foundation of the entire argument. It connects the physical concept of persistence to the mathematical concept of invariance, setting the stage for the algebraic structures that invariance brings with it.

3.3 From Invariants to Algebraic Structure

We now show that the collection of invariants of a persistent system is not merely a set—it necessarily carries rich algebraic structure.

Proposition 3.5 (Invariants Form an Algebra). *Let G be a group acting on a set X , and let R be a commutative ring with unity (for example, $R = \mathbb{R}$). The set of G -invariant functions $\text{Inv}_G(X, R) = \{f : X \rightarrow R \mid f(g \cdot x) = f(x) \text{ for all } g \in G, x \in X\}$ forms a subalgebra of the algebra of all R -valued functions on X . Specifically, $\text{Inv}_G(X, R)$ is closed under:*

- (a) **Addition:** *If $f, h \in \text{Inv}_G(X, R)$, then $f + h \in \text{Inv}_G(X, R)$.*
- (b) **Scalar multiplication:** *If $f \in \text{Inv}_G(X, R)$ and $r \in R$, then $r \cdot f \in \text{Inv}_G(X, R)$.*
- (c) **Pointwise multiplication:** *If $f, h \in \text{Inv}_G(X, R)$, then $f \cdot h \in \text{Inv}_G(X, R)$.*
- (d) **Constants:** *Every constant function is in $\text{Inv}_G(X, R)$.*

Proof. For (a): let $f, h \in \text{Inv}_G(X, R)$ and $g \in G$. Then $(f + h)(g \cdot x) = f(g \cdot x) + h(g \cdot x) = f(x) + h(x) = (f + h)(x)$. Parts (b)–(d) follow by analogous direct computations. $\square$

This proposition establishes that the invariants of a persistent system do not float in isolation—they form a structured algebraic object. One can add, subtract, and multiply conserved quantities and obtain new conserved quantities. In physics, this is a familiar fact: if energy E and momentum p are both conserved, then so is any polynomial combination of E and p .

The significance for our argument is that this algebraic structure arises *automatically* from the requirement of persistence. We did not impose algebraic structure on the system from outside; it emerged as a logical consequence of the system’s having invariants at all.

Proposition 3.6 (Invariants Encode Group Structure). *Let G act on X , and suppose the invariant algebra $\text{Inv}_G(X, \mathbb{R})$ separates the orbits of G (that is, for any two distinct orbits $G \cdot x \neq G \cdot y$, there exists $f \in \text{Inv}_G(X, \mathbb{R})$ with $f(x) \neq f(y)$). Then the algebraic structure of $\text{Inv}_G(X, \mathbb{R})$ uniquely determines the orbit structure of the G -action, and hence encodes the symmetry group G up to the equivalence relevant for the action.*

Proof sketch. The orbit space X/G is the quotient of X by the equivalence relation $x \sim y$ if and only if $y = g \cdot x$ for some $g \in G$. The invariant algebra $\text{Inv}_G(X, \mathbb{R})$ factors through this

quotient: every invariant function is constant on orbits, and the separation hypothesis ensures that distinct orbits are distinguished by invariants. Therefore $\text{Inv}_G(X, \mathbb{R})$ is isomorphic (as an algebra) to the algebra of functions on the orbit space, and this orbit space encodes the “shape” of the G -action. By classical invariant theory [Weyl, 1946, Hilbert, 1893], when G is a reductive group acting on a vector space, the invariant algebra is finitely generated and its structure determines the quotient variety $X//G$, which captures the essential geometry of the action. $\square$

The upshot of Proposition 3.6 is that the invariants of a persistent system do not merely form an algebra—they encode the system’s symmetries. Knowing the conserved quantities is equivalent to knowing the symmetry group, and vice versa (this is the content of Noether’s theorem, now seen from the algebraic side). The persistent system therefore carries, as part of its essential structure, a definite group-theoretic architecture.

3.4 Grounding the Persistence Filter: Why Do Physical Systems Have Invariants?

A natural objection to the preceding development is that it risks circularity: we define persistence in terms of invariant observables, and then note that invariant observables carry algebraic structure. Is the persistence filter doing genuine explanatory work, or is it merely relabeling the explanandum—re-describing “why does mathematics work?” as “why are there conserved quantities?”

This objection deserves a careful response, because the question it raises—*why do physical systems possess non-trivial conserved quantities in the first place?*—is the deepest stratum of Wigner’s puzzle. We offer three complementary answers.

First: the dynamical answer. The existence of conserved quantities in Hamiltonian and Lagrangian systems is not a contingent empirical finding but a mathematical consequence of the structure of the dynamics itself. In any system governed by a smooth Lagrangian, Noether’s theorem (Theorem 2.8) guarantees that continuous symmetries produce conservation laws. The question then reduces to: why does the dynamics have symmetries? For fundamental physics, the answer lies in the homogeneity and isotropy of spacetime—properties that, far from being arbitrary, are among the weakest possible regularity conditions on a physical arena. A spacetime that lacked translation invariance would be one in which the laws of physics differed from point to point with no pattern whatsoever; the absence of even approximate symmetry would amount to a kind of maximal disorder that itself requires explanation [Earman, 1989]. Symmetry, in this sense, is the default; it is perfect asymmetry that would be “unreasonable.”

Second: the thermodynamic answer. Statistical mechanics provides a complementary perspective. Among the vast space of possible microstates, configurations with conserved macroscopic quantities overwhelmingly dominate: the number of microstates compatible with a given value of total energy vastly exceeds the number of microstates that happen to cluster around any other macroscopic variable. This is the content of the maximum entropy principle [Jaynes, 1957]. Systems with conservation laws occupy overwhelmingly larger regions of phase space than systems without them. The “filter” is therefore statistical: random sampling from the space of possible configurations overwhelmingly yields configurations

with conserved quantities, not because a filter selectively removes non-conserving systems, but because such systems constitute a set of measure zero.

Third: the logical answer. The persistence filter is not circular even taken on its own terms, because it operates at a different level of explanation from the theories it produces. The filter does not *explain* any specific conservation law (energy, charge, baryon number); rather, it explains why the existence of *some* conservation laws is a precondition for the existence of systems stable enough to be observed at all. This is analogous to the role of natural selection in biology: selection does not explain any specific adaptation, but it explains why adapted organisms are the ones we find. The analogy is precise: just as selection is a filter on the space of phenotypes that favors fitness-enhancing traits without specifying which traits those will be, the persistence filter is a filter on the space of dynamical configurations that favors invariant-possessing systems without specifying which invariants those will be. The specific invariants are determined by the specific dynamics; the persistence filter explains why invariants in general are ubiquitous. (An important disanalogy should be noted: natural selection operates on a population of entities that exist and compete, while the persistence filter operates on the space of *conceivable* configurations, most of which were never instantiated. The filter is therefore closer to a constraint on what is possible than to a selective process acting on what exists. It does not explain why these systems exist rather than others—only that if they exist and persist, they must have invariants.)

These three answers are mutually reinforcing, but they apply to different classes of systems and should not be conflated. For fundamental Lagrangian systems (particle physics, classical mechanics, general relativity), the dynamical answer is primary: the symmetries of spacetime and the Lagrangian structure directly produce exact conservation laws via Noether’s theorem. For macroscopic physical systems (thermodynamic systems, many-body systems, continuum mechanics), the thermodynamic answer is primary: even when exact microscopic symmetries are broken or inaccessible, the statistical dominance of conservation-compatible microstates ensures the emergence of approximate macroscopic invariants. For non-physical domains—biological homeostasis, economic equilibria, social institutions—the logical answer provides the appropriate justification: these systems persist because feedback mechanisms, evolutionary selection, or institutional enforcement maintain certain quantities within viable ranges, and the persistence filter explains why such invariant-possessing systems are the ones available for scientific investigation. The dynamical answer explains why fundamental physical systems have exact symmetries. The thermodynamic answer explains why macroscopic systems generically exhibit approximate invariants. The logical answer explains why the existence of invariants is not circular but structurally necessary for any system that can be an object of scientific investigation.

4 The Main Result

We now assemble the components developed in the preceding sections into the central argument of the paper.

4.1 The Invariant Structure Theorem

Theorem 4.1 (Invariant Structure Theorem). *Let (X, T, Φ) be an exactly persistent dynamical system with persistent region $A \subseteq X$ and a non-trivial collection $\mathcal{F}$ of conserved observables. Let $G = \{\Phi_t \mid t \in T\}$ be the group (or monoid) of time-evolution maps. Then:*

(i) *The collection $\mathcal{F}$ generates a commutative $\mathbb{R}$ -algebra $\mathfrak{A} = \langle \mathcal{F} \rangle_{\mathbb{R}}$ of G -invariant functions on A .*

(ii) *The algebra $\mathfrak{A}$ defines an equivalence relation $\sim_{\mathfrak{A}}$ on A by:*

$$x \sim_{\mathfrak{A}} y \iff f(x) = f(y) \text{ for all } f \in \mathfrak{A}. \quad (13)$$

(iii) *The quotient $A/\sim_{\mathfrak{A}}$ inherits from $\mathfrak{A}$ the structure of an affine algebraic set (in the finite-dimensional case) or a topological algebra spectrum (in general).*

(iv) *This quotient structure is a model of a first-order theory $\mathcal{T}_{\mathfrak{A}}$ whose language includes function symbols for the algebra operations and relation symbols for the ordering and metric structures inherited from $\mathbb{R}$.*

Proof. Part (i): By Proposition 3.5, the G -invariant functions form a subalgebra of all real-valued functions on A . The algebra generated by $\mathcal{F}$ consists of all finite polynomial combinations of elements of $\mathcal{F}$ with real coefficients, and is contained in $\text{Inv}_G(A, \mathbb{R})$ since each generator is invariant.

Part (ii): The relation $\sim_{\mathfrak{A}}$ is clearly an equivalence relation (reflexive, symmetric, transitive). It identifies points that are indistinguishable by the invariant observables—these are points that, as far as the conserved quantities can discern, are in the same physical state.

Part (iii): When $\mathfrak{A}$ is finitely generated as an $\mathbb{R}$ -algebra, say $\mathfrak{A} = \mathbb{R}[f_1, \dots, f_k]$, the map $\pi : A \rightarrow \mathbb{R}^k$ defined by $\pi(x) = (f_1(x), \dots, f_k(x))$ is a well-defined map whose image $\pi(A)$ is a subset of $\mathbb{R}^k$. The quotient $A/\sim_{\mathfrak{A}}$ is homeomorphic (in the quotient topology) to $\pi(A)$, which inherits the structure of a semi-algebraic set. In the general (infinite-dimensional) case, the quotient is identified with the Gelfand spectrum of $\mathfrak{A}$ —the space of maximal ideals—equipped with the Zariski or weak-* topology, following standard commutative algebra [Atiyah and MacDonald, 1969].

Part (iv): The quotient $A/\sim_{\mathfrak{A}}$, equipped with the operations inherited from $\mathfrak{A}$ (addition, multiplication, scalar multiplication) and the ordering and metric from $\mathbb{R}$, constitutes a structure in the model-theoretic sense (Definition 2.10). We define the first-order language $\mathcal{L}_{\mathfrak{A}}$ to have:

- constant symbols for the distinguished elements 0 and 1;
- binary function symbols for $+$ and $\times$;
- a unary function symbol for each generator f_i ;
- binary relation symbols for $=$ and $\leq$.

The theory $\mathcal{T}_{\mathfrak{A}}$ consists of the axioms of commutative ordered $\mathbb{R}$ -algebras together with any additional identities satisfied by the generators. The quotient $A/\sim_{\mathfrak{A}}$ is, by construction, a model of $\mathcal{T}_{\mathfrak{A}}$. $\square$

This theorem is the mathematical core of our argument. It shows that the invariant structure of a persistent system is not an amorphous collection of unrelated facts, but a *structured algebraic object* that is, by the very nature of its construction, a model of a definite mathematical theory. The physical system does not need to be “translated into” mathematics from outside—the mathematics is already there, encoded in the system’s invariant structure.

4.2 The Persistence Filter Theorem

We now state the main result, which synthesizes the preceding developments into a single theorem.

Theorem 4.2 (Persistence Filter Theorem). *Let $\mathfrak{U}$ be the class of all dynamical systems (X, T, Φ) , and let $\mathfrak{P} \subset \mathfrak{U}$ be the subclass of persistent systems (those admitting a non-trivial persistent region A with at least one conserved observable). Then:*

- (i) *Every system in $\mathfrak{P}$ possesses a non-trivial algebra of invariants $\mathfrak{A}$.*
- (ii) *Every such algebra $\mathfrak{A}$ defines a first-order theory $\mathcal{T}_{\mathfrak{A}}$ of which the system (quotiented by observational equivalence) is a model.*
- (iii) *Therefore, every persistent system admits a faithful mathematical description: there exists a first-order theory $\mathcal{T}$ and a structure-preserving map (a model-theoretic interpretation) from $\mathcal{T}$ to the system’s invariant structure.*

*We call the passage from $\mathfrak{U}$ to $\mathfrak{P}$ the **persistence filter**: only systems with the structural regularity required for mathematical describability survive it.*

Proof. Part (i) follows from Lemma 3.4: persistence requires invariants, and invariants generate an algebra (Proposition 3.5). Part (ii) follows from Theorem 4.1(iv): the invariant algebra defines a first-order theory of which the system’s quotient structure is a model. Part (iii) is the conjunction of (i) and (ii).

The “filter” metaphor is justified as follows: consider a dynamical system with no invariants whatsoever. By Lemma 3.4, such a system has no conserved observables and hence no persistent region—it is not persistent. It does not appear in $\mathfrak{P}$. The systems that do appear in $\mathfrak{P}$ are precisely those with enough invariant structure to be models of non-trivial mathematical theories. $\square$

The Persistence Filter Theorem provides our answer to Wigner’s puzzle. Mathematics is effective in describing the physical world not because of a mysterious pre-established harmony, but because:

1. The physical world consists of systems that persist through time.
2. Persistence requires invariants.
3. Invariants necessarily carry algebraic structure.
4. This algebraic structure is rich enough to be a model of a substantive mathematical theory (Theorem 4.4 below).

5. Therefore, the physical world is necessarily described by formal mathematical theories with genuine predictive content.

Each step in this chain is a theorem, not an assumption. The conclusion follows by the transitivity of logical implication.

4.3 Structural Richness: Beyond Trivial Describability

A careful reader will have noticed that one step in our argument—the observation that any structure with definite properties is a model of *some* first-order theory—is, taken in isolation, near-trivial. Given any structure $\mathcal{M}$, one can always define a theory $\text{Th}(\mathcal{M})$ consisting of all first-order sentences true in $\mathcal{M}$; then $\mathcal{M}$ is a model of $\text{Th}(\mathcal{M})$ by construction. This “diagram theory” is not the kind of deep, predictive mathematics that prompted Wigner’s astonishment. The real content of our argument must therefore lie in showing that persistent systems are models of theories that are *substantive*—finitely axiomatizable, possessed of non-trivial consequences, and shared across diverse physical instantiations.

This subsection provides that content.

Definition 4.3 (Structural Richness). Let $\mathcal{M}$ be a model of a first-order theory $\mathcal{T}$. We say $\mathcal{T}$ is **structurally rich** relative to $\mathcal{M}$ if:

- (a) **Finite axiomatizability:** $\mathcal{T}$ is axiomatized by a finite set of axioms (or axiom schemata with finitely many parameters).
- (b) **Non-trivial consequence:** $\mathcal{T}$ has consequences not contained in the quantifier-free diagram of $\mathcal{M}$ —that is, $\mathcal{T}$ entails universal or existential statements that are not simply records of individual facts about $\mathcal{M}$.
- (c) **Multiple realizability:** $\mathcal{T}$ has models that are not isomorphic to $\mathcal{M}$ —the theory captures a *type* of structure, not merely a single instance.

These three conditions are jointly sufficient to distinguish “Wigner-type” mathematical describability from mere logical tautology. A finitely axiomatizable theory is learnable and communicable. A theory with non-trivial consequences can generate predictions. A multiply-realizable theory unifies diverse phenomena under a common structure. These are precisely the features of the mathematical theories that Wigner found unreasonably effective.

Theorem 4.4 (Structural Richness of Persistent Systems). *Let (X, T, Φ) be a persistent dynamical system whose invariant algebra is finitely generated: $\mathfrak{A} = \mathbb{R}[\phi_1, \dots, \phi_k]$ for finitely many conserved observables $\phi_1, \dots, \phi_k$. Let $\mathcal{I} \subset \mathbb{R}[x_1, \dots, x_k]$ be the ideal of polynomial relations satisfied by the generators:*

$$\mathcal{I} = \{p \in \mathbb{R}[x_1, \dots, x_k] \mid p(\phi_1, \dots, \phi_k) = 0 \text{ on } A\}. \quad (14)$$

Then the first-order theory $\mathcal{T}_{\mathfrak{A}}$ associated with the system (Theorem 4.1) is structurally rich:

- (a) $\mathcal{T}_{\mathfrak{A}}$ is finitely axiomatizable: it is axiomatized by the axioms of commutative $\mathbb{R}$ -algebras together with a finite set of polynomial identities generating $\mathcal{I}$.

- (b) $\mathcal{T}_{\mathfrak{A}}$ has non-trivial consequences: any identity $p(\phi_1, \dots, \phi_k) = 0$ with $p \in \mathcal{I}$ constrains the joint values of the conserved observables, ruling out configurations that are algebraically possible but dynamically forbidden.
- (c) $\mathcal{T}_{\mathfrak{A}}$ is multiply realizable: the same polynomial ideal $\mathcal{I}$ can be satisfied by different physical systems with different state spaces, provided they share the same algebraic relations among their conserved quantities.

Proof. Part (a): By the Hilbert Basis Theorem [Hilbert, 1893], the ideal $\mathcal{I}$ is finitely generated: $\mathcal{I} = (p_1, \dots, p_m)$ for some finite set of polynomials. The theory $\mathcal{T}_{\mathfrak{A}}$ is therefore axiomatized by the (finitely many) axioms of commutative $\mathbb{R}$ -algebras plus the m sentences $\forall \bar{x} (p_j(\bar{x}) = 0)$ for $j = 1, \dots, m$.

Part (b): Suppose $\mathcal{I}$ contains a non-trivial relation $p(\phi_1, \dots, \phi_k) = 0$. This relation constrains the values that the conserved observables can jointly assume: the image of the map $(\phi_1, \dots, \phi_k) : A \rightarrow \mathbb{R}^k$ lies in the algebraic variety $V(\mathcal{I}) = \{\bar{x} \in \mathbb{R}^k \mid p_j(\bar{x}) = 0, j = 1, \dots, m\}$, which is a proper subset of $\mathbb{R}^k$ whenever $\mathcal{I} \neq \{0\}$. This constitutes a non-trivial universal prediction: any state of the system must have its conserved quantities lying on $V(\mathcal{I})$.

Part (c): The variety $V(\mathcal{I})$ is a geometric object defined purely by polynomial equations. Different physical systems—with different state spaces, different dynamics, and different physical interpretations—can realize the same variety if their conserved quantities satisfy the same polynomial relations. For instance, the relation $E^2 = p^2 c^2 + m^2 c^4$ is satisfied by every relativistic particle regardless of its specific identity, spin, or interaction type. Each such particle system is a distinct model of the same finitely axiomatized theory. $\square$

This theorem closes the gap between mere logical describability and substantive mathematical physics. The algebraic variety $V(\mathcal{I})$ is a concrete, non-trivial geometric object that constrains the system’s behavior, unifies diverse physical instantiations, and generates testable predictions. This is the kind of mathematical structure whose effectiveness demands explanation, and it arises not by coincidence but by the structural necessity of the persistence filter.

Remark 4.5 (Scope of the Finite-Generation Hypothesis). Theorem 4.4 assumes that the invariant algebra $\mathfrak{A}$ is finitely generated. This assumption holds automatically in two important cases: (i) when the symmetry group G is a reductive algebraic group acting on a finite-dimensional vector space (by the Hilbert–Nagata theorem [Hilbert, 1893, Weyl, 1946]), which covers the symmetry groups of fundamental physics (the Lorentz group, compact gauge groups, etc.); and (ii) when the system has finitely many independent conserved quantities, as is the case for finite-dimensional Hamiltonian systems with n degrees of freedom (which have at most $2n - 1$ independent first integrals).

For systems where finite generation may fail—notably infinite-dimensional systems, biological systems with hierarchical regulatory structure, and social systems with evolving institutional rules—the theorem does not apply directly. In such cases, the invariant algebra may be infinitely generated, and the Hilbert Basis Theorem does not guarantee finite axiomatizability. The Gelfand spectrum construction of Theorem 4.1(iii) remains available: the system is still described by the topological algebra spectrum of $\mathfrak{A}$, but this description may not be finitely axiomatizable.

For these domains, the appropriate analogue of structural richness is that the invariant algebra, though potentially infinite-dimensional, admits finite-dimensional *approximations* of controlled quality—precisely the (ε, τ) -approximate models of Section 12.2. The practical mathematical models used in biology and the social sciences are, on this view, finite-dimensional approximations to a potentially infinite invariant structure, and the quality of the approximation is bounded by the persistence parameters. We regard the development of structural richness criteria for infinitely generated invariant algebras as an important open problem.

Proposition 4.6 (The Depth Gradient). *Let (X, T, Φ) be an (ε, τ) -persistent system (Definition 3.2) with k independent (ε, τ) -approximate invariants $\phi_1, \dots, \phi_k$ satisfying m approximate polynomial relations of maximum degree d . Define the **algebraic richness** of the system as the triple $\rho = (k, m, d)$ and the **model precision** as the approximation bound δ of Proposition 12.4. Then:*

- (a) *The model precision δ is bounded by a polynomial in ε of degree at most d : $\delta \leq C(k, m, d) \cdot \varepsilon$, where C depends on the algebraic richness but not on the specific system.*
- (b) *The number of independent testable predictions generated by the approximate theory is at least m (one per independent polynomial relation).*
- (c) *Across any class of systems sharing the same algebraic type but differing in persistence quality ε , the precision of mathematical description degrades monotonically with ε .*

This proposition formalises the empirical observation that mathematical precision decreases systematically as one moves from fundamental physics (many exact invariants, $\varepsilon \approx 0$, rich polynomial relations) through the life sciences (fewer, approximate invariants, moderate ε) to the social sciences (few fragile invariants, large ε). The gradient is not a post hoc rationalisation but a quantitative prediction of the PFT: the precision of available mathematical models is bounded by the persistence parameters of the systems they describe. Fundamental physics occupies the high end of this gradient because fundamental symmetry groups are large and conservation laws hold to extraordinary precision. Biological and social systems occupy lower positions because their invariants are maintained by feedback and enforcement mechanisms that are inherently approximate.

4.4 The Functorial Perspective

The relationship between persistent physical systems and their mathematical descriptions can be given a precise categorical formulation that clarifies the sense in which the word “isomorphism” is used in the title of this paper.

Definition 4.7 (Categories of Persistent Systems and Formal Theories). Define two categories:

- **Pers**: the category whose objects are persistent dynamical systems $(X, T, \Phi, \mathcal{F})$ (with designated conserved observables) and whose morphisms are equivariant maps $h : X_1 \rightarrow X_2$ satisfying $h \circ \Phi_t^1 = \Phi_t^2 \circ h$ and $\phi_2 \circ h = \phi_1$ for corresponding observables.

- **Form**: the category whose objects are first-order theories $\mathcal{T}$ and whose morphisms are theory interpretations (translations of the language of one theory into another that preserve provability).

Theorem 4.8 (Functorial Correspondence). *The assignment $\mathfrak{J} : \mathbf{Pers}^{\text{op}} \rightarrow \mathbf{Form}$ defined by*

$$\mathfrak{J}(X, T, \Phi, \mathcal{F}) = \mathcal{T}_{\mathfrak{A}}, \quad (15)$$

where $\mathfrak{A} = \langle \mathcal{F} \rangle_{\mathbb{R}}$ is the invariant algebra and $\mathcal{T}_{\mathfrak{A}}$ is the first-order theory it defines (Theorem 4.1), is a contravariant functor from $\mathbf{Pers}$ to $\mathbf{Form}$ (equivalently, a covariant functor from the opposite category $\mathbf{Pers}^{\text{op}}$). That is, it reverses the direction of morphisms: an equivariant map $h : \mathcal{S}_1 \rightarrow \mathcal{S}_2$ between persistent systems induces an interpretation $\mathfrak{J}(h) : \mathcal{T}_{\mathfrak{A}_2} \rightarrow \mathcal{T}_{\mathfrak{A}_1}$ between theories, and this assignment respects identities and composition:

$$\mathfrak{J}(\text{id}) = \text{id}, \quad (16)$$

$$\mathfrak{J}(h_2 \circ h_1) = \mathfrak{J}(h_1) \circ \mathfrak{J}(h_2). \quad (17)$$

Proof sketch. An equivariant map $h : (X_1, T, \Phi^1, \mathcal{F}_1) \rightarrow (X_2, T, \Phi^2, \mathcal{F}_2)$ induces a pullback $h^* : \mathfrak{A}_2 \rightarrow \mathfrak{A}_1$ on invariant algebras (by $h^*(f) = f \circ h$), which in turn induces an interpretation $\mathfrak{J}(h) : \mathcal{T}_{\mathfrak{A}_2} \rightarrow \mathcal{T}_{\mathfrak{A}_1}$ (contravariantly, following the standard duality between algebra homomorphisms and geometric morphisms [Hodges, 1993]). The contravariance of the pullback— $(h_2 \circ h_1)^* = h_1^* \circ h_2^*$ —ensures that Equation (17) holds. $\square$

The contravariant functor $\mathfrak{J}$ provides the precise sense in which the title’s use of “isomorphism” should be understood. We do not claim that each physical system is literally identical to a mathematical object—that would be a category error. Rather, there is a *systematic, structure-preserving, contravariant correspondence* between the category of persistent physical systems and a subcategory of formal theories. Physical relationships between systems (equivariant maps) correspond contravariantly to logical relationships between theories (interpretations): a map “from system 1 to system 2” yields a translation “from the theory of system 2 to the theory of system 1.” This duality is familiar from algebraic geometry (where morphisms of varieties correspond contravariantly to homomorphisms of coordinate rings) and reflects a deep structural pattern. The correspondence is functorial—it respects the compositional structure on both sides—and this functoriality, rather than literal isomorphism, is the precise content of the title’s claim.

Remark 4.9 (On the Strength of “Isomorphism”). The functor $\mathfrak{J}$ is not, in general, full, faithful, or essentially surjective—so it does not establish an equivalence of categories. The title’s reference to “isomorphism” should be understood not in the technical categorical sense of an equivalence, but in the broader scientific sense of a structure-preserving correspondence: every persistent physical system maps to a substantive mathematical theory, this mapping respects the morphism structure on both sides, and the resulting theories have genuine content (Theorem 4.4). Establishing conditions under which $\mathfrak{J}$ (or a suitable restriction) is full or faithful remains an open problem.

5 Why These Structures? The Specificity Problem

The argument so far establishes that persistent systems are necessarily describable by *some* mathematical theory. But Wigner’s puzzle has a sharper edge: why do *specific* mathematical structures—Lie groups, differential equations, Hilbert spaces, Riemannian manifolds—recur so persistently across diverse areas of physics? This section addresses the specificity problem.

5.1 Continuity and Differentiable Structure

Proposition 5.1 (Persistence Selects for Continuity). *Let (X, T, Φ) be a dynamical system where X is a topological space, $T = \mathbb{R}$, and $A \subseteq X$ is an (ε, τ) -persistent region with respect to a family $\mathcal{F}$ of continuous observables. Then the restriction $\Phi|_{\mathbb{R} \times A} : \mathbb{R} \times A \rightarrow X$ is continuous in t for each fixed $x \in A$.*

Proof. Fix $x \in A$ and a continuous observable $\phi \in \mathcal{F}$. The persistence condition (Definition 3.2) requires $\|\phi(\Phi(t, x)) - \phi(x)\| < \varepsilon$ for all $t \in [0, \tau]$. Now consider the function $\psi_x : \mathbb{R} \rightarrow V$ defined by $\psi_x(t) = \phi(\Phi(t, x))$. Suppose, for contradiction, that ψ_x has a discontinuity at some $t_0 \in [0, \tau)$: there exists $\delta > 0$ such that for all $\eta > 0$, there is a t' with $|t' - t_0| < \eta$ and $\|\psi_x(t') - \psi_x(t_0)\| \geq \delta$.

By the semigroup property (Equation (2)), $\psi_x(t') = \phi(\Phi(t', x)) = \phi(\Phi(t' - t_0, \Phi(t_0, x)))$. Setting $y = \Phi(t_0, x) \in X$ and $s = t' - t_0$, we have $\|\phi(\Phi(s, y)) - \phi(y)\| \geq \delta$ for $|s|$ arbitrarily small. If $y \in A$ (which holds when A is forward-invariant under Φ), this contradicts (ε, τ) -persistence for any $\varepsilon < \delta$.

Therefore ψ_x is continuous for every continuous observable $\phi \in \mathcal{F}$. If $\mathcal{F}$ separates points of X (or generates the topology of X), then continuity of all ψ_x implies continuity of $t \mapsto \Phi(t, x)$ in the weak topology induced by $\mathcal{F}$. $\square$

An important clarification is in order. This proposition does not derive topological structure from nothing; it *presupposes* that the state space X carries a topology and that the relevant observables are continuous. The claim is not that persistence creates topology *ex nihilo*, but rather that among dynamical systems on a topological state space, persistence selects for continuous evolution. The topological structure of the state space itself may reflect deeper facts about the system (such as the continuum structure of physical spacetime) or, as explored in Section 9, features of the cognitive apparatus that constructs the model.

This is significant because continuity (and, by extension, differentiability, when the evolution is smooth) is not merely a mathematical convenience—it is a *consequence of persistence on topological spaces*. And once we have differentiable evolution, the entire apparatus of differential equations, Lie groups, and differential geometry becomes applicable, not by fiat, but by structural necessity.

5.2 Symmetry Groups and Representation Theory

Proposition 5.2 (Persistence Selects for Symmetry). *Let (X, T, Φ) be a persistent system with conserved observables $\mathcal{F} = \{\phi_1, \dots, \phi_k\}$. Define the symmetry group of the persistent regime as:*

$$G_{\mathcal{F}} = \{g \in \text{Diff}(X) \mid \phi_i \circ g = \phi_i \text{ for all } i = 1, \dots, k\}, \quad (18)$$

where $\text{Diff}(X)$ denotes the group of diffeomorphisms of X (or homeomorphisms, if X is not smooth). Then $G_{\mathcal{F}}$ is a group, and the time-evolution maps $\{\Phi_t\}_{t \in T}$ form a one-parameter subgroup of $G_{\mathcal{F}}$.

This proposition shows that persistence does not merely imply *some* symmetry—it implies a structured symmetry group that contains the system’s dynamics as a subgroup. The group $G_{\mathcal{F}}$ is the largest group of transformations under which all conserved quantities are invariant. It is this group that organizes the system’s behavior and makes it predictable.

The reason certain symmetry groups (rotation groups, Lorentz groups, gauge groups) recur across physics is that they arise from the most general forms of persistence under the most basic transformations—spatial rotations, changes of reference frame, and changes of internal phase. These are not arbitrary choices; they are the symmetries that *must* be present if the laws of physics are to be the same everywhere, at all times, and for all observers.

5.3 Variational Principles and Optimality

A striking feature of fundamental physics is that its laws can be expressed as variational principles: the actual trajectory of a system extremizes (usually minimizes) a functional called the action. This is not merely an elegant reformulation; it has deep structural significance.

Proposition 5.3 (Persistence and Variational Stability). *Let (X, T, Φ) be a persistent dynamical system whose evolution is governed by a Lagrangian L . Let $\gamma_0 : [t_1, t_2] \rightarrow X$ be a trajectory lying in the persistent region A , and let γ_η be a one-parameter family of perturbed trajectories with $\gamma_0 = \gamma_\eta|_{\eta=0}$ and fixed endpoints $\gamma_\eta(t_1) = \gamma_0(t_1)$, $\gamma_\eta(t_2) = \gamma_0(t_2)$. If γ_0 is persistent in the sense that small perturbations do not cause the trajectory to leave A , then γ_0 is a critical point of the action functional $S[\gamma] = \int_{t_1}^{t_2} L(\gamma(t), \dot{\gamma}(t), t) dt$.*

Proof. Persistent trajectories, by definition, satisfy $\|\phi(\gamma_\eta(t)) - \phi(\gamma_0(t))\| < \varepsilon$ for all conserved observables ϕ , all $t \in [t_1, t_2]$, and all sufficiently small $|\eta|$. In particular, the action $S[\gamma_\eta]$ is a conserved observable of the extended system (trajectory space with its natural dynamics), and persistence requires $|S[\gamma_\eta] - S[\gamma_0]| < \varepsilon$ for small $|\eta|$.

Now expand $S[\gamma_\eta]$ to second order in η :

$$S[\gamma_\eta] = S[\gamma_0] + \eta \left. \frac{dS}{d\eta} \right|_{\eta=0} + \frac{\eta^2}{2} \left. \frac{d^2 S}{d\eta^2} \right|_{\eta=0} + O(\eta^3). \quad (19)$$

If $\left. \frac{dS}{d\eta} \right|_{\eta=0} \neq 0$, then for small η of appropriate sign, $|S[\gamma_\eta] - S[\gamma_0]|$ can be made arbitrarily large relative to η , violating the bounded-deviation condition for any fixed ε . (Formally: given $\varepsilon > 0$, set $\eta = 2\varepsilon / \left| \left. \frac{dS}{d\eta} \right|_{\eta=0} \right|$ with the appropriate sign; then $|S[\gamma_\eta] - S[\gamma_0]| \geq 2\varepsilon - O(\eta^2)$, which exceeds ε for sufficiently small η .)

Therefore $\left. \frac{dS}{d\eta} \right|_{\eta=0} = 0$, which is the defining condition for γ_0 to be a critical point of S . $\square$

Two clarifications are essential. First, this argument applies to Lagrangian systems—it requires that the dynamics be derivable from an action principle. Not all dynamical systems admit a Lagrangian formulation (dissipative systems generally do not), and for such systems the variational characterization does not apply directly, though analogous stability

conditions exist (e.g., Lyapunov stability criteria). Second, the argument establishes that persistent trajectories are critical points, but not all critical points are persistent—saddle points of the action are critical but unstable. Persistence corresponds to critical points that are also *stable* (local minima or centers), adding a condition beyond mere criticality.

The broader point remains: the variational structure of physics is not an arbitrary mathematical convenience but a reflection of the persistence filter. Among all conceivable trajectories, those that persist are the stable critical points of the action, and this stability is what the Euler–Lagrange equations express.

5.4 Hilbert Spaces and Linearity

The appearance of Hilbert spaces and linear operators in quantum mechanics has been considered particularly mysterious. Why should the state space of nature be a complex vector space with an inner product?

The persistence filter offers a partial—but more rigorous than often appreciated—answer. We develop it in two steps.

Proposition 5.4 (Persistence of Quantum States Requires Unitarity). *Let $\mathcal{H}$ be the state space of a quantum system, equipped with a notion of transition probability $P(\psi, \phi) = |\langle \psi | \phi \rangle|^2$ between states. If the system is persistent in the sense that transition probabilities are invariant under time evolution— $P(\psi(t), \phi(t)) = P(\psi(0), \phi(0))$ for all t and all state pairs—then the time evolution operator $U(t)$ is either unitary or anti-unitary. If $t \mapsto U(t)$ is continuous, then $U(t)$ is unitary.*

Proof. This is the content of Wigner’s theorem [Wigner, 1931]: any bijection on the set of rays (one-dimensional subspaces) of a Hilbert space that preserves transition probabilities is implemented by a unitary or anti-unitary operator. The requirement of continuity in t eliminates the anti-unitary option (anti-unitary operators cannot form a continuous one-parameter group), leaving $U(t)$ unitary. $\square$

Proposition 5.5 (Unitary Evolution Requires Hilbert Space Structure). *A one-parameter group of unitary operators $\{U(t)\}_{t \in \mathbb{R}}$ that is strongly continuous (i.e., $\lim_{t \rightarrow 0} U(t)\psi = \psi$ for all ψ) acts on a complex Hilbert space and, by Stone’s theorem [Stone, 1932], takes the form $U(t) = e^{-iHt}$ for a self-adjoint operator H (the Hamiltonian).*

These two propositions together show that the Hilbert space structure of quantum mechanics is not an arbitrary mathematical choice but a consequence of requiring that quantum systems persist (in the sense that their measurable transition probabilities remain stable over time). The argument proceeds: persistence of measurement outcomes $\Rightarrow$ invariance of transition probabilities $\Rightarrow$ Wigner’s theorem $\Rightarrow$ unitary evolution $\Rightarrow$ Stone’s theorem $\Rightarrow$ Hilbert space with self-adjoint Hamiltonian.

This derivation is not complete—it presupposes the existence of a probabilistic structure on the state space and does not derive the complex number field from first principles. These remain open problems (see Section 12.3). Nevertheless, the argument shows that the core mathematical apparatus of quantum mechanics (Hilbert space, unitary operators, self-adjoint Hamiltonians) follows from the persistence of statistical regularities, providing a concrete

instance of the persistence filter at work in the most mathematically sophisticated domain of physics.

Moreover, the superposition principle—that quantum states can be added and scaled—is connected to persistence through linearity. If two states are individually persistent (stable energy eigenstates), their coherent superposition is also persistent provided the dynamics are linear. Weinberg [Weinberg, 1989] showed that even small nonlinear corrections to quantum mechanics would lead to superluminal signaling and violations of the second law of thermodynamics—that is, to the rapid destruction of the macroscopic persistence on which all observation depends. The filter selects for linear dynamics because nonlinear quantum evolution would generically destroy the coherence required for persistence of superposition states.

6 Illustrative Models

The abstract chain of reasoning developed in Sections 3 to 5 benefits from concrete illustration. This section presents five worked examples spanning discrete, continuous, conservative, dissipative, and biological systems. Each example constructs the invariant algebra explicitly and demonstrates the persistence filter in action.

6.1 A Minimal Discrete System

Consider the simplest illustration of the persistence filter: a finite dynamical system.

Example 6.1 (Cyclic Permutation vs. Orbit Decomposition). Let $X = \{1, 2, 3, 4, 5, 6\}$, $T = \mathbb{Z}$, and $\Phi_1 : X \rightarrow X$ be the cyclic permutation $\Phi_1 = (1\ 2\ 3\ 4\ 5\ 6)$. This permutation visits every state, so every orbit is the entire set X . The only Φ_1 -invariant functions $f : X \rightarrow \mathbb{R}$ are the constants: $\text{Inv}_{\langle \Phi_1 \rangle}(X, \mathbb{R}) = \mathbb{R}$. The invariant algebra is trivial—it encodes no non-trivial structure.

Now consider instead $\Phi_2 = (1\ 2\ 3)(4\ 5)(6)$, which decomposes X into three orbits: $\{1, 2, 3\}$, $\{4, 5\}$, and $\{6\}$. The invariant algebra $\text{Inv}_{\langle \Phi_2 \rangle}(X, \mathbb{R})$ consists of all functions constant on orbits. Choosing characteristic functions χ_A, χ_B, χ_C for the three orbits, we have $\text{Inv} = \mathbb{R}[\chi_A, \chi_B, \chi_C]/(\chi_A + \chi_B + \chi_C - 1, \chi_A^2 - \chi_A, \chi_B^2 - \chi_B, \chi_C^2 - \chi_C)$, a three-dimensional algebra isomorphic to $\mathbb{R}^3$.

The contrast is sharp: the system with a single orbit (no persistent substructure) has a trivial invariant algebra. The system with multiple orbits (persistent substates) has a non-trivial algebra whose dimension equals the number of orbits. The invariant algebra detects and encodes the persistent decomposition.

6.2 Double Pendulum: Persistence Versus Chaos

The double pendulum provides a striking physical illustration of how the invariant algebra “thins out” as persistence degrades.

At low energies, the double pendulum oscillates quasi-periodically. The individual arm energies E_1 and E_2 are approximately conserved (they exchange energy slowly), so the approximate invariant algebra is $\mathbb{R}[E, E_1] \cong \mathbb{R}[E, E_2]$ (since $E = E_1 + E_2$ is exactly conserved).

This algebra is two-dimensional, the motion is confined to a two-torus in phase space, and the system admits an effective two-degree-of-freedom mathematical description via the KAM theorem [Arnold, 1963].

At high energies, the motion becomes chaotic. The approximate conservation of E_1 and E_2 breaks down—their fluctuations become comparable to their mean values—and the only remaining exact invariant is the total energy E . The invariant algebra collapses from $\mathbb{R}[E, E_1]$ to $\mathbb{R}[E]$ —from a two-dimensional algebra to a one-dimensional one. The system’s mathematical tractability degrades in precise lockstep with the loss of invariants: in the chaotic regime, the motion fills a three-dimensional energy surface rather than a two-dimensional torus, and long-term prediction becomes exponentially sensitive to initial conditions.

This example illustrates Proposition 4.6: the richness of the mathematical description is directly controlled by the richness of the invariant algebra, which is in turn controlled by the degree of persistence.

6.3 Cellular Automata: Persistence Without Noether

Conway’s Game of Life [Gardner, 1970] is a discrete, deterministic, non-Lagrangian dynamical system—there is no Lagrangian, no continuous symmetry, and no Noether theorem. Yet persistent structures abound.

A *glider* is a configuration of five cells that translates diagonally by one cell every four time steps, returning to its original shape. The persistent observables of a glider include its velocity $v = (1, 1)/4$ (cells per step), its period $\tau = 4$, and its shape (up to translation). These observables are exactly conserved under the dynamics (until the glider interacts with another structure).

The invariant algebra of an isolated glider is generated by v and τ —both constants—and by the shape indicator function σ (which takes value 1 on the glider’s footprint and 0 elsewhere, invariant under time-translation by 4 steps). This is a small but non-trivial algebra: $\mathbb{R}[v_x, v_y, \tau, \sigma]$ with the relations $v_x = v_y = 1/4$, $\tau = 4$, $\sigma^2 = \sigma$.

By contrast, a random initial configuration in the Game of Life typically dissolves into a chaotic transient before (if ever) settling into a collection of persistent structures (still lifes and oscillators). During the transient, no non-trivial observables are conserved, and the system resists mathematical description. After the transient, the surviving persistent structures each carry their own invariant algebra.

This example demonstrates that the persistence filter operates in discrete, non-Lagrangian settings, answering the concern that the thesis relies too heavily on continuous symmetry and Noether’s theorem.

6.4 A Dissipative System: The Lorenz Attractor

The Lorenz system [Lorenz, 1963]

$$\dot{x} = \sigma(y - x), \quad \dot{y} = x(\rho - z) - y, \quad \dot{z} = xy - \beta z \quad (20)$$

(with standard parameters $\sigma = 10$, $\rho = 28$, $\beta = 8/3$) is dissipative: phase-space volumes contract exponentially, and there is no Hamiltonian or Lagrangian formulation. Nevertheless,

the Lorenz attractor is a persistent structure—trajectories on the attractor remain on it indefinitely.

The invariants of the Lorenz attractor are not conserved quantities in the Noether sense but rather geometric and statistical properties: the correlation dimension $d_2 \approx 2.05$, the largest Lyapunov exponent $\lambda_1 \approx 0.91$, and the invariant (Sinai–Ruelle–Bowen) measure μ_{SRB} . These quantities are invariant under time evolution on the attractor and under smooth coordinate changes. They generate a (function-algebraic) structure: for instance, the generalized dimensions d_q for different q satisfy functional relations derivable from the thermodynamic formalism [Beck and Schlögl, 1993].

The mathematical theory of the Lorenz attractor—its topological classification (a branched manifold), its symbolic dynamics (a shift on a specific subshift of finite type), its ergodic properties—constitutes a substantive, predictive mathematical description of a persistent dissipative structure, derived entirely from the attractor’s invariant properties.

6.5 Biological Homeostasis: A Thermoregulation Model

Consider a minimal model of biological thermoregulation:

$$\frac{dT}{dt} = -\alpha(T - T_{\text{set}}) + \eta(t), \quad \frac{dT_{\text{set}}}{dt} = 0, \quad (21)$$

where T is body temperature, T_{set} is the homeostatic setpoint, $\alpha > 0$ is the feedback gain, and $\eta(t)$ represents environmental perturbations.

The setpoint T_{set} is an exact conserved quantity—it does not change under the dynamics. The time-averaged temperature $\langle T \rangle$ converges to T_{set} on the attractor (when η is mean-zero). The feedback gain α is a parameter invariant under time evolution.

The invariant algebra is $\mathbb{R}[T_{\text{set}}, \alpha]$ —a two-dimensional commutative algebra. The associated first-order theory includes the axioms $T_{\text{set}} > 0$, $\alpha > 0$, and the relation $\langle T \rangle = T_{\text{set}}$ (on the attractor). This is a non-trivial, finitely axiomatized, multiply realizable theory: it applies equally to mammalian thermoregulation, to bacterial chemotaxis (with “concentration setpoint” replacing “temperature setpoint”), and to engineering thermostats.

This example makes concrete the claim of Section 8.1: biological persistence is maintained by feedback mechanisms that generate conserved quantities (setpoints), and these conserved quantities carry algebraic structure amenable to mathematical description. The mathematical tractability of homeostatic systems is a direct consequence of the persistence filter.

7 Comparison with Existing Approaches

7.1 Mathematical Universe Hypothesis (Tegmark)

Max Tegmark’s Mathematical Universe Hypothesis (MUH) [Tegmark, 2008] asserts that external physical reality *is* a mathematical structure—not merely described by one, but identical to one. Every consistent mathematical structure exists physically, and our universe is one such structure.

The Persistence Filter Thesis differs from the MUH in several important respects. First, the MUH is an *ontological* claim (the world is made of math), while the PFT is an *epistemological and structural* claim (the world is describable by math because of the conditions for persistence). Second, the MUH faces the well-known “measure problem”—it must explain why we observe this particular mathematical structure and not another, leading to the need for a measure over the space of all structures. The PFT avoids this problem by not claiming that all mathematical structures are physically real; instead, it claims only that all persistent physical structures are mathematically describable. Third, the MUH treats mathematical describability as a brute fact (the world just *is* math), while the PFT provides a *mechanism*—the persistence filter—that explains why the describability holds.

7.2 Structural Realism

Structural realism, developed by Worrall, Ladyman, French, and others [Worrall, 1989, Ladyman and Ross, 2007, French, 2014], holds that what is preserved across theory change in science is structure rather than ontology. An important distinction within the structural realist camp separates *epistemic* structural realism (ESR), which claims that all we can *know* about the world is its structure, from *ontic* structural realism (OSR), which claims that structure is all there *is*—that the world is constituted by relations and structures rather than by objects bearing those relations [Ladyman and Ross, 2007].

The Persistence Filter Thesis is compatible with and provides a foundation for both versions, though it aligns more naturally with OSR. The invariant algebra $\mathfrak{A}$ of a persistent system *is* the structure that structural realism claims is preserved across theory change. Under ESR, $\mathfrak{A}$ represents the structural knowledge we can extract about the system; under OSR, $\mathfrak{A}$ *constitutes* the system’s identity. Our argument shows *why* this structure exists (because of the requirements of persistence) and *why* it is mathematical (because invariant algebras are models of formal theories). In this sense, the PFT provides a *mechanism* for the structural realist’s foundational claim, grounding it not in metaphysical stipulation but in dynamical necessity.

The PFT also engages with the “mapping account” of mathematical applicability developed by Pincock [Pincock, 2012] and by Bueno and French [Bueno and French, 2018]. On this account, mathematics is applicable to the physical world when there exists a suitable mapping (homomorphism, partial isomorphism, or structural similarity) between a mathematical structure and a physical domain. The PFT explains *why* such mappings exist: the persistence filter guarantees that physical systems carry algebraic structure, and the functor $\mathfrak{J}$ (Theorem 4.8) provides the systematic mapping that the account requires. The contribution of the PFT is to show that the existence of suitable mappings is not a brute fact but a consequence of dynamical persistence.

7.3 Selection Bias and Anthropic Arguments

The selection bias response to Wigner—that we notice mathematical successes and ignore failures—has merit but is incomplete. It explains why we might *overestimate* the effectiveness of mathematics, but it does not explain why mathematics works as well as it does in

fundamental physics, where the precision of agreement between theory and experiment can reach parts per billion (as in the anomalous magnetic moment of the electron).

The Persistence Filter Thesis subsumes the selection bias argument and goes further. Yes, there is a selection effect—but it is not merely cognitive (we notice successes). It is *physical*: only persistent structures exist to be observed. The selection is not in our attention but in reality itself.

7.4 Evolutionary and Cognitive Accounts

Evolutionary accounts [Dennett, 1995] argue that natural selection equipped humans with mathematical abilities because those abilities helped our ancestors survive. This explains why we can do simple arithmetic and geometry, but it does not explain why the same mathematical structures that describe everyday mechanics also describe quantum field theory, general relativity, and the behavior of neutron stars.

The PFT complements evolutionary accounts by locating the source of mathematical effectiveness not in the human mind but in the physical world. Evolution explains why we can *discover* mathematical structure; the persistence filter explains why the structure is *there to be discovered*. De Cruz [De Cruz, 2006] offered a more nuanced version of the evolutionary account, arguing that mathematical practices constitute cognitive scaffolding that amplifies innate capacities far beyond their evolutionary function. This is a plausible account of mathematical *capacity*, but it does not explain the *correspondence*—why cultural mathematical constructions happen to match the structure of the physical world. The PFT provides the missing half of the explanation.

7.5 Steiner’s Tripartite Distinction

Steiner [Steiner, 1998] sharpened Wigner’s puzzle by distinguishing three forms of mathematical applicability: *semantic* (how abstract entities describe concrete ones), *descriptive* (why mathematical language is adequate for physical description), and *heuristic* (why mathematical analogies guide the discovery of new physics). The heuristic applicability—exemplified by Dirac’s prediction of the positron by trusting the mathematics of his equation—is the most resistant to explanation and the strongest evidence for what Steiner called “Pythagoreanism.”

The PFT addresses all three forms. Semantic applicability is explained by the functorial correspondence (Theorem 4.8): the invariant algebra provides the concrete-to-abstract mapping. Descriptive applicability follows from the Persistence Filter Theorem (Theorem 4.2): every persistent system is a model of a substantive theory. Heuristic applicability receives a partial explanation from the Structural Richness Theorem (Theorem 4.4): because the invariant algebra is multiply realizable, mathematical structures discovered in one physical context can be transferred to another—the *same* algebra can be realized by different systems, so mathematical analogy reflects genuine structural kinship.

7.6 Conventionalism

A persistent tradition, originating with Poincaré [Poincaré, 1902] and Duhem [Duhem, 1906], holds that the “fit” between mathematics and physics is partly an artifact of how theories are constructed. The specific mathematical formulation involves conventional choices—coordinate systems, units, gauge conventions, regularization schemes—and the precision of the description is partly a consequence of how the theory is built.

The PFT does not deny the role of convention. What it denies is that the *entire* mathematical content of physics is conventional. The invariant algebra $\mathfrak{A}$ (Theorem 4.1) is defined by the dynamics, not by the theorist’s choices—it is the structure that remains after all conventional choices have been factored out. The PFT thus provides a precise criterion for distinguishing the conventional from the non-conventional: the invariant algebra is objective; its particular presentation (choice of generators, coordinates, formalism) is conventional.

7.7 Information-Theoretic and Computational Approaches

Wheeler’s “it from bit” program [Wheeler, 1990] proposed that physical reality is fundamentally informational. Müller and Masanes [Müller and Masanes, 2013] showed that quantum mechanics’ Hilbert space structure can be derived from information-theoretic postulates—the existence of continuous reversible transformations between pure states, a maximally mixed state, and purifiability. This remarkable result addresses the specificity problem for quantum mechanics.

The PFT provides a complementary perspective: the information-theoretic postulates can be understood as persistence conditions—requirements that must be satisfied by any quantum system that persists as a carrier of information. Deutsch and Marletto’s constructor theory [Marletto and Deutsch, 2015] offers a related framework, replacing dynamical laws with statements about which transformations are possible and which are impossible. A system persists in constructor-theoretic terms if it can function as a constructor—an entity that causes a transformation and remains able to do so again. The invariants of such a system are the possible and impossible tasks, which carry algebraic structure (the structure of a monoid of possible transformations). Constructor theory thus provides a natural alternative formalization of the persistence filter independent of the Lagrangian formalism.

8 Applications Beyond Physics

The Persistence Filter Thesis is not limited to fundamental physics. Any domain in which systems persist through time is a domain in which the thesis applies. This section explores implications for biology, the social sciences, and the foundations of mathematics.

8.1 Biological Systems

Living organisms are paradigmatic persistent systems. An organism maintains its structural and functional integrity over time through homeostatic mechanisms—feedback loops that keep temperature, pH, blood glucose, and countless other variables within viable ranges.

The PFT predicts that persistent biological systems will possess conserved quantities and symmetries, and that these will be amenable to mathematical description. This prediction is confirmed by the extensive use of mathematical models in biology: the Hodgkin–Huxley equations for neural dynamics [Hodgkin and Huxley, 1952], the Lotka–Volterra equations for population dynamics, the Hardy–Weinberg equilibrium in population genetics [Hardy, 1908], and the extensive use of information-theoretic quantities (entropy, mutual information) in computational neuroscience and genomics.

The persistence filter explains why these mathematical models work: the biological systems they describe are persistent, and their persistence is maintained by conserved quantities (homeostatic setpoints, genetic information, population-level statistical regularities) that necessarily carry algebraic structure.

8.2 Social and Economic Systems

Persistent social institutions—legal systems, markets, languages, cultural norms—also possess invariant structure that admits mathematical description. The persistence of a market economy, for example, requires the conservation of certain accounting identities (assets equal liabilities plus equity), the stability of price mechanisms, and the enforcement of contract law. These conserved relations form the algebraic backbone on which mathematical economics is built.

The PFT predicts that the more persistent a social institution is, the more amenable it will be to mathematical modeling—and conversely, that institutions in rapid flux or transition will be less well-described by stable mathematical frameworks. This prediction aligns with the empirical observation that equilibrium economics succeeds best in describing stable markets, while crises and transitions require different, often less mathematically tractable, approaches.

8.3 The Foundations of Mathematics Itself

The PFT has implications for the philosophy of mathematics. If mathematical structures are the invariant skeletons of persistent systems, then mathematical truth has a natural interpretation: a mathematical statement is true if it holds in the invariant structure of every persistent system that realizes the relevant algebraic type. This is a form of *structuralism* about mathematical truth [Shapiro, 1997], but grounded not in abstract metaphysics but in the physical conditions for persistence.

This also suggests a naturalistic account of mathematical discovery: mathematicians discover the structural features that any persistent system of a given type must possess. The reason that pure mathematical research so often produces structures that later find physical application is that both the mathematician and the physical world are, in a sense, exploring the same space—the space of structures compatible with persistence.

9 The Kantian Counter-Thesis: Mathematics as a Coordinate Grid of Consciousness

The preceding sections argued that mathematics describes the world because the world’s persistent structures necessarily carry algebraic architecture. There is, however, a radical alternative that must be confronted with equal rigor: perhaps mathematics does not reflect the external world at all. Perhaps it is instead a *coordinate grid imposed by consciousness*—a scaffolding that the mind projects onto an internal model of the environment, which is itself a construction of the mind. On this view, the “unreasonable effectiveness” dissolves because there is no gap to bridge: the structures we discover in nature are the very structures our cognitive apparatus uses to *construct* what we call nature.

This section develops the alternative as a formal thesis, demonstrates its internal consistency and explanatory power, then marshals equally rigorous arguments against it, and finally draws conclusions from the confrontation.

9.1 The Thesis Stated and Defended

We begin by formalizing the cognitive-grid hypothesis within the same mathematical vocabulary used throughout this paper. The aim is to show that this hypothesis is not a vague philosophical gesture but a logically coherent position with precise consequences.

Definition 9.1 (Cognitive Modeling System). A **cognitive modeling system** is a quadruple $(\Sigma, \mathcal{M}_\Sigma, \Pi, W)$, where:

- Σ is a *sensory input space*—the set of all possible patterns of stimulation that the organism’s sensory apparatus can register (neural activation patterns, receptor states, etc.);
- $\mathcal{M}_\Sigma$ is the *internal model space*—a structured set that the cognitive system constructs and maintains, representing the organism’s “world-model”;
- $\Pi : \Sigma \rightarrow \mathcal{M}_\Sigma$ is the *projection operator*—the map by which raw sensory input is organized, compressed, and embedded into the internal model;
- W is the *external environment*—the mind-independent reality (whatever it may be) that causally impinges on Σ .

The crucial observation is that the organism never has direct access to W . All it has access to is Σ and, through the projection Π , the model $\mathcal{M}_\Sigma$. The “world” as experienced and investigated by the organism is $\mathcal{M}_\Sigma$, not W .

Definition 9.2 (Cognitive Coordinate Structure). The **cognitive coordinate structure** is the collection of structural constraints that Π imposes on $\mathcal{M}_\Sigma$. Formally, let $\mathcal{C} = \{C_1, C_2, \dots\}$ be a family of relations and operations on $\mathcal{M}_\Sigma$ that are determined by Π rather than by W . The cognitive coordinate structure is the relational system $(\mathcal{M}_\Sigma, \mathcal{C})$.

The cognitive-grid hypothesis can now be stated precisely:

Proposition 9.3 (Cognitive Grid Thesis — Formal Statement). *Let $(\Sigma, \mathcal{M}_\Sigma, \Pi, W)$ be a cognitive modeling system. Suppose the projection operator Π factors through a structured representation:*

$$\Pi = \iota \circ \rho, \quad \rho : \Sigma \rightarrow \mathcal{R}, \quad \iota : \mathcal{R} \hookrightarrow \mathcal{M}_\Sigma, \quad (22)$$

where $\mathcal{R}$ is a representation space equipped with algebraic structure (a vector space, a group action, a metric space, etc.) that is determined by the architecture of the cognitive system, not by properties of W . Then the mathematical structure observed in $\mathcal{M}_\Sigma$ reflects the structure of $\mathcal{R}$ —that is, of the cognitive apparatus—rather than the structure of W .

The factorization in Equation (22) encodes a specific claim: the raw sensory data Σ is first compressed and restructured by ρ into a format dictated by the organism’s neural architecture, and only then is it embedded into the world-model. The algebraic structure of $\mathcal{R}$ (linearity, metric properties, symmetries) is contributed by the cognitive system, not extracted from the environment.

We now develop three arguments in support of this thesis.

Argument 1: The Encoding Theorem. The structure of any representation is jointly determined by source and channel. Information theory makes this precise.

Proposition 9.4 (Structural Imprint of the Channel). *Let $W \xrightarrow{\sigma} \Sigma \xrightarrow{\rho} \mathcal{R}$ be a chain of maps, where σ represents causal impingement of the environment on the sensorium and ρ represents cognitive encoding. Define the effective structure $\mathcal{S}_{\text{eff}}$ on $\mathcal{R}$ as the image of the composite map $\rho \circ \sigma$. Then $\mathcal{S}_{\text{eff}}$ is constrained not only by the structure of W but also by the structure of ρ —specifically, by $\ker(\rho)$ (the equivalence classes ρ conflates) and by the algebraic properties ρ preserves or introduces.*

If ρ is a linear map (as in many neural coding models), then $\mathcal{S}_{\text{eff}}$ is a linear subspace of $\mathcal{R}$ —regardless of whether W has linear structure.

Proof. Let $V = \text{im}(\rho \circ \sigma) \subseteq \mathcal{R}$. If ρ is linear, then for any $w_1, w_2 \in W$ and scalar α :

$$\rho(\alpha \cdot \sigma(w_1) + \sigma(w_2)) = \alpha \cdot \rho(\sigma(w_1)) + \rho(\sigma(w_2)). \quad (23)$$

Thus, whenever $\sigma(w_1)$ and $\sigma(w_2)$ are in the domain of ρ , their images satisfy the vector space axioms in $\mathcal{R}$, even if the underlying elements $w_1, w_2 \in W$ do not form a vector space. The linear structure of V is inherited from ρ , not from W . $\square$

This is a rigorous demonstration that the cognitive encoding channel can *introduce* mathematical structure that is not present in the source. The linearity that we observe in our models may be a property of how we encode, not of what we encode.

Argument 2: The Kantian Consistency Theorem. We show that the cognitive-grid hypothesis is fully consistent with all observational evidence.

Proposition 9.5 (Observational Indistinguishability). (This and subsequent results in Section 9 are formalized philosophical arguments: the definitions and logical structure are rigorous, but the premises encode philosophical positions whose acceptance is not compelled by mathematics alone.)

Let W be an external environment and let $(\Sigma, \mathcal{M}_\Sigma, \Pi, W)$ be a cognitive modeling system. Define two hypotheses:

- H_{ext} : The mathematical structure in $\mathcal{M}_\Sigma$ faithfully reflects structure in W .
- H_{cog} : The mathematical structure in $\mathcal{M}_\Sigma$ is imposed by Π and does not necessarily reflect the structure of W .

Then H_{ext} and H_{cog} are observationally indistinguishable: for any observation $o \in \mathcal{M}_\Sigma$, both hypotheses predict o .

Proof. Every observation available to the organism is an element of $\mathcal{M}_\Sigma = \text{im}(\Pi)$. Under H_{ext} , the structure of $\mathcal{M}_\Sigma$ matches W because Π faithfully transmits it. Under H_{cog} , the structure of $\mathcal{M}_\Sigma$ is what it is because Π constructs it. In both cases, the organism observes the same $\mathcal{M}_\Sigma$ with the same structure. Since the organism has no access to W except through Π , no measurement within $\mathcal{M}_\Sigma$ can distinguish between the two hypotheses. $\square$

This theorem is the formal core of the Kantian position. It establishes that the question “does mathematical structure reside in the world or in our representation of the world?” cannot be settled by any empirical observation, because every observation is already filtered through the representational system whose contribution is precisely what is at issue.

Argument 3: Neuroscientific Precedent. The claim that cognitive architecture imposes structure on experience has extensive empirical support. Color perception provides a well-studied example. The trichromatic structure of human color vision—the fact that all perceived colors can be represented as points in a three-dimensional space—is a property of the human visual system (three types of cone cells), not of the electromagnetic spectrum (which is infinite-dimensional). Other species with different photoreceptor systems perceive different color spaces. The three-dimensionality of perceived color is a coordinate grid imposed by biology [Helmholtz, 1867].

Analogously, the cognitive-grid thesis proposes that the mathematical structures we discover in physics—three-dimensional Euclidean space, continuous time, smooth dynamics—may be features of our representational architecture rather than mind-independent features of reality. Kant proposed precisely this for space and time; modern neuroscience suggests the same may hold for more abstract mathematical structures.

9.2 The Counter-Thesis: Refutation of the Pure Cognitive Grid

We now argue, with equal rigor, that the cognitive-grid hypothesis *cannot* be the complete explanation for the effectiveness of mathematics. The refutation proceeds through four independent arguments, each identifying a feature of mathematical physics that the cognitive-grid thesis cannot accommodate.

Argument 1: The Prediction Asymmetry.

Proposition 9.6 (Predictive Excess of Mathematical Physics). *Let $(\Sigma, \mathcal{M}_\Sigma, \Pi, W)$ be a cognitive modeling system, and let $\mathcal{T}$ be a mathematical theory constructed within $\mathcal{M}_\Sigma$. If the structure of $\mathcal{T}$ is entirely determined by Π (the cognitive-grid hypothesis), then $\mathcal{T}$ can encode regularities of Π but cannot systematically generate correct predictions about novel sensory inputs $\sigma_{\text{new}} \in \Sigma$ that were not part of the data used to construct $\mathcal{T}$.*

Proof by contrapositive. Suppose $\mathcal{T}$ generates a prediction $p \in \mathcal{M}_\Sigma$ about a domain of sensory input not yet encountered. Suppose this prediction is subsequently confirmed: the new sensory input, when projected by Π , matches p . Under H_{cog} , the structure of $\mathcal{T}$ reflects only Π , not W . For the prediction to succeed, one of two conditions must hold:

- (a) The new sensory input σ_{new} happens, by coincidence, to be compatible with the structure Π imposes. For a single prediction, this is possible. For systematic, repeated, high-precision success across diverse domains, the probability of coincidence becomes vanishingly small.
- (b) The structure that Π imposes is itself *adapted to* regularities in W —that is, Π has been shaped (by evolution, development, or learning) so that its structural properties track structural properties of W . But this concedes that W possesses structure, and that the structure in $\mathcal{M}_\Sigma$ is not purely cognitive but partly reflects W .

In either case, purely cognitive structure is insufficient to explain systematic predictive success. $\square$

The force of this argument lies in the historical record. Dirac’s equation, constructed from purely mathematical considerations of relativistic invariance and spinor structure, predicted the existence of the positron—an entirely novel entity that no one had observed or imagined. The mathematical formalism of general relativity predicted gravitational lensing, frame-dragging, and gravitational waves decades before they were observed. These are not cases of the mind recognizing patterns it has imposed; they are cases of mathematical structure revealing features of reality that the mind had not anticipated. A purely cognitive grid could not systematically generate such surprises.

Argument 2: The Intersubjectivity Constraint.

Proposition 9.7 (Convergence Across Distinct Cognitive Architectures). *Let $(\Sigma_1, \mathcal{M}_1, \Pi_1, W)$ and $(\Sigma_2, \mathcal{M}_2, \Pi_2, W)$ be two cognitive modeling systems with different architectures (different sensory apparatuses Σ_i , different projection operators Π_i , different model spaces $\mathcal{M}_i$). If the mathematical structure in each model space is purely a product of the respective cognitive architecture, then there is no reason to expect $\mathcal{M}_1$ and $\mathcal{M}_2$ to converge on the same mathematical description of a shared domain.*

Yet convergence is precisely what we observe. The laws of mechanics were developed independently by Newton and Leibniz using different formalisms (fluxions vs. infinitesimals), yet they arrived at equivalent mathematical content. Heisenberg’s matrix mechanics and Schrödinger’s wave mechanics were developed from entirely different conceptual starting points, yet von Neumann proved them mathematically equivalent [von Neumann, 1932]. Different cognitive systems, using different representational strategies, converge on the same structure. This convergence is expected if the structure originates in W (a common external source) but anomalous if it originates solely in Π (since different Π ’s should produce different structures).

Argument 3: The Rigidity of Mathematical Constraint.

Proposition 9.8 (Non-Malleability of Physical Law). *If mathematical structure in physics were a cognitive projection, then it would be expected to exhibit the flexibility characteristic of cognitive constructions: susceptibility to framing effects, cultural variation, and revision by stipulation. Physical laws exhibit the opposite: they are rigid, culturally invariant, and resistant to revision by fiat.*

No amount of cognitive effort allows an engineer to stipulate that the gravitational constant has a different value, or that energy is not conserved, or that the speed of light in vacuum depends on the observer’s velocity. These constraints feel external precisely because they are—they resist the very cognitive flexibility that would characterize a purely projected structure. A coordinate grid imposed by consciousness should be, at least in principle, modifiable by consciousness. The mathematical structure of physics is not.

Argument 4: The Formal Independence Theorem.

Proposition 9.9 (Cognitive Architecture Does Not Determine Physical Theory). *Let Π be a fixed cognitive projection operator. The space of first-order theories consistent with the structural constraints imposed by Π is strictly larger than the space of physical theories confirmed by observation. That is, if $\mathcal{T}_\Pi$ denotes the set of theories compatible with the cognitive architecture and $\mathcal{T}_{\text{phys}}$ denotes the set of theories confirmed by experiment, then $\mathcal{T}_{\text{phys}} \subsetneq \mathcal{T}_\Pi$.*

Proof. The cognitive architecture Π constrains the *form* of theories (e.g., they must be expressible in terms of continuous quantities if Π produces a continuous model space). However, within these formal constraints, many distinct theories are possible. For example, the cognitive constraint of spatial continuity is compatible with Newtonian gravity ($F \propto r^{-2}$), Coulomb’s law ($F \propto r^{-2}$), a hypothetical r^{-3} force law, and infinitely many others. The fact that $F \propto r^{-2}$ holds for gravity while r^{-3} does not is not determined by Π —it is determined by W . Therefore, the selection of the actual physical theory from among all theories compatible with cognitive architecture requires input from W , and the cognitive grid alone is explanatorily insufficient. $\square$

This theorem identifies the precise locus of insufficiency: the cognitive grid may constrain the *language* of physics (continuous, quantitative, spatially structured), but it cannot determine the *content* (which equations, which constants, which symmetry groups). The content comes from the world.

9.3 Synthesis: The Dual-Source Structure of Mathematical Physics

The confrontation between the cognitive-grid thesis and its refutation yields a result more nuanced and more informative than either thesis alone. We formalize the synthesis.

Proposition 9.10 (Dual-Source Decomposition). *Let $(\Sigma, \mathcal{M}_\Sigma, \Pi, W)$ be a cognitive modeling system, and let $\mathcal{T}_{\text{phys}}$ be the mathematical theory that the organism constructs to describe its experience. Then $\mathcal{T}_{\text{phys}}$ admits a decomposition:*

$$\mathcal{T}_{\text{phys}} = \mathcal{T}_{\text{form}} \cup \mathcal{T}_{\text{cont}}, \quad (24)$$

where:

- $\mathcal{T}_{\text{form}}$ is the formal component—the structural constraints contributed by the cognitive architecture Π (continuity, dimensionality, linearity of the representational space, the language of differential equations, etc.);
- $\mathcal{T}_{\text{cont}}$ is the content component—the specific laws, constants, and symmetries that are determined by the structure of W as transmitted through Σ .

The two components are not independent: $\mathcal{T}_{\text{form}}$ constrains the admissible elements of $\mathcal{T}_{\text{cont}}$, and $\mathcal{T}_{\text{cont}}$ is non-trivial (not fully determined by $\mathcal{T}_{\text{form}}$) whenever W possesses structure that is not already encoded in Π .

Proof. The existence of $\mathcal{T}_{\text{form}}$ follows from Proposition 9.4: the encoding channel Π imposes structural constraints that appear as axioms or background assumptions of $\mathcal{T}_{\text{phys}}$. The existence and non-triviality of $\mathcal{T}_{\text{cont}}$ follows from Proposition 9.9: cognitive architecture does not determine physical theory uniquely, so additional content must come from empirical input reflecting W .

To show that $\mathcal{T}_{\text{form}}$ constrains $\mathcal{T}_{\text{cont}}$: any specific law in $\mathcal{T}_{\text{cont}}$ must be expressible in the language afforded by $\mathcal{T}_{\text{form}}$. A cognitive system that represents the world through continuous functions cannot formulate theories that require intrinsically discrete primitive operations (though it can approximate them).

To show that $\mathcal{T}_{\text{cont}}$ is non-trivial: by Proposition 9.9, the set of theories compatible with $\mathcal{T}_{\text{form}}$ strictly contains $\mathcal{T}_{\text{phys}}$. The selection principle—what picks out $\mathcal{T}_{\text{phys}}$ from $\mathcal{T}_{\text{form}}$ —is empirical input from W , encoded in $\mathcal{T}_{\text{cont}}$. $\square$

This decomposition has several important consequences.

First, it dissolves the apparent dichotomy between “mathematics reflects the world” and “mathematics is a cognitive projection.” Both claims are partially correct, but neither is complete. Mathematics as we practice it is the product of a *coupling* between cognitive structure and world structure, and disentangling the two contributions is a non-trivial (and perhaps not always possible) task.

Second, it recontextualizes the Persistence Filter Thesis. The PFT, developed in Section 4, argued that persistent systems necessarily carry mathematical structure. The Dual-Source Decomposition adds precision: the *type* of mathematical structure we discover (continuous vs. discrete, linear vs. nonlinear, finite-dimensional vs. infinite-dimensional) is partly determined by our cognitive architecture ($\mathcal{T}_{\text{form}}$), while the *specific content* of that structure (which symmetry groups, which conservation laws, which coupling constants) is determined by the persistent structures in W ($\mathcal{T}_{\text{cont}}$). The persistence filter operates on W , but what passes through it is refracted by Π before we can apprehend it.

Third, the synthesis explains a puzzling asymmetry in the history of science: while the general *framework* of mathematical physics (calculus, linear algebra, differential geometry) has remained stable across centuries and cultures—suggesting a cognitive-architectural origin—the *specific theories* within that framework (from Ptolemaic epicycles to Newtonian gravity to general relativity) have undergone revolutionary change driven by empirical discovery—confirming the role of external structure. The Dual-Source Decomposition predicts exactly this pattern: $\mathcal{T}_{\text{form}}$ is stable (it reflects the relatively fixed cognitive architecture of *Homo sapiens*), while $\mathcal{T}_{\text{cont}}$ evolves (as we gain deeper empirical access to W).

Remark 9.11 (Relation to Kant). The Dual-Source Decomposition is, in a precise sense, a mathematization of Kant’s transcendental philosophy [Kant, 1781]. Kant argued that space, time, and causality are *forms of intuition*—structures that the mind contributes to experience, not properties extracted from things-in-themselves. Our $\mathcal{T}_{\text{form}}$ corresponds to Kant’s forms of intuition, and our $\mathcal{T}_{\text{cont}}$ corresponds to the empirical content that fills those forms. The advance beyond Kant is twofold: we have formalized the distinction in model-theoretic terms, and we have shown—through the Persistence Filter Thesis—that the empirical content $\mathcal{T}_{\text{cont}}$ is itself constrained by a structural necessity (persistence), so that the world’s contribution is not arbitrary but lawful.

Remark 9.12 (Implications for Artificial Intelligence). If different cognitive architectures contribute different $\mathcal{T}_{\text{form}}$ components, then artificial cognitive systems—with architectures radically different from biological neural networks—may develop different mathematical formalisms for describing the same physical world. The content $\mathcal{T}_{\text{cont}}$ should be shared (since both systems interact with the same W), but the formal scaffolding may diverge. This is a testable prediction: as AI systems develop increasingly autonomous scientific reasoning capabilities, the mathematical languages they produce may differ from human mathematics in form while agreeing in empirical content.

10 Counterarguments and Responses

A thesis of this scope must confront its strongest objections. This section identifies nine counterarguments—including those raised by reviewers and anticipated critics—and provides responses that clarify the boundaries and commitments of the Persistence Filter Thesis.

10.1 The Central Argument Is Vacuously True

Objection. Any structure with definite properties is a model of some first-order theory—one can always write down the theory whose axioms are the true sentences of that structure. If the conclusion holds for every structure, persistent or not, then the persistence filter is doing no explanatory work, and the thesis reduces to a tautology.

Response. The objection correctly identifies that the mere existence of *some* formal theory is not what demands explanation. We concede that Theorem 4.2, taken in isolation, does not fully resolve the puzzle; its contribution is that the mathematical theory guaranteed to exist is not an arbitrary list of truths but one with algebraic structure (Proposition 3.5), finite axiomatizability, non-trivial polynomial constraints, and multiple realizability (Theorem 4.4). These properties distinguish the invariant algebra of a persistent system from a mere catalogue of facts. We acknowledge that a gap remains between “possesses algebraic structure” and “admits deep, predictive mathematical physics,” and we regard closing this gap fully as an open problem (Section 12.3) rather than a defeater for the thesis.

10.2 Circularity in the Definition of Persistence

Objection. Persistence is defined in terms of stable observables, and the conclusion is that persistent systems possess stable observables. The argument appears circular: the

definiendum restates the definiens.

Response. The definition of persistence is indeed given in terms of stable observables—this is a stipulation, not a theorem. The substantive content lies in the *consequences*: that invariants necessarily form a structured algebra (Proposition 3.5), that this algebra encodes the symmetry group (Proposition 3.6), and that the resulting structure is a model of a finitely axiomatizable theory (Theorem 4.4). None of these consequences are contained in the definition. The deeper question—why physical systems possess non-trivial conserved quantities—is addressed in Section 3.4 through dynamical, thermodynamic, and logical arguments.

10.3 Noether’s Theorem Does Not Generalize as Claimed

Objection. The paper relies on Noether’s theorem for the symmetry-conservation connection, but this theorem applies only to continuous symmetries of Lagrangian systems. The extensions to biology, economics, and social systems lack Lagrangian formulations.

Response. This is a fair criticism, addressed in the revised Section 2.3. The core argument (Sections 3 and 4) does not depend on Noether’s theorem. It requires only that persistent systems possess at least one non-trivial conserved observable, regardless of the mechanism producing it—Noether charge, discrete quantum number, topological invariant, Lyapunov function, or homeostatic setpoint. The illustrative models of Section 6 (cellular automata, the Lorenz attractor, the thermoregulation model) demonstrate the argument in non-Lagrangian settings.

10.4 The Thesis Is Unfalsifiable

Objection. If a system persists, the thesis says it has mathematical structure. If it fails to persist, the filter excludes it. No conceivable observation could refute the thesis, disqualifying it as a scientific claim.

Response. The thesis is a structural meta-thesis about the relationship between persistence and describability, not a physical theory competing with specific models. The appropriate standard is not strict Popperian falsifiability but explanatory adequacy and the generation of testable auxiliary predictions. Section 11 identifies six such predictions, several of which are testable with current tools. A systematic failure of the predicted correlation between persistence and mathematical tractability, or the routine construction of predictive models for systems with no identifiable invariants, would constitute evidence against the thesis.

10.5 The Specificity Problem Is Not Resolved

Objection. The thesis explains that persistent systems must possess *some* algebraic structure, but not why *these particular* structures—Lie groups, Riemannian manifolds, Hilbert spaces—recur across fundamental physics.

Response. We accept this criticism as partly valid. The specificity arguments of Section 5 identify structural mechanisms—persistence selects for continuity (Proposition 5.1), variational stability (Proposition 5.3), and unitarity (Proposition 5.4)—but these arguments

are individually suggestive rather than definitive. The general form of each argument is: “persistence under transformations of type X requires invariance structure of type Y , and Y is the defining property of mathematical framework Z .” We maintain that this substantially narrows the class of admissible structures, even if a complete derivation of the specific mathematical frameworks of fundamental physics from persistence conditions alone remains open.

10.6 The Thesis Reduces to Structural Realism

Objection. The PFT may be a restatement of ontic structural realism rather than a new resolution of Wigner’s puzzle.

Response. The PFT overlaps with structural realism but is not reducible to it. Structural realism asserts that the world has mathematical structure but does not explain *why*. The PFT provides a mechanism—the persistence filter—and generates auxiliary predictions (the correlation between persistence and tractability, the structural transformation of invariant algebras at phase transitions) that structural realism, as a general philosophical position, does not.

10.7 The Biological and Social Extensions Are Too Speculative

Objection. The extensions of Section 8 to biology, economics, and social institutions are asserted rather than demonstrated. Biological homeostasis involves active feedback, not passive Hamiltonian invariance.

Response. The extensions are programmatic. However, the algebraic consequences of Proposition 3.5 depend only on the *existence* of invariant observables, not on the mechanism maintaining them. Whether an invariant is conserved by Lagrangian symmetry, feedback control, or institutional enforcement, the algebraic closure properties are identical. Section 6.5 provides a worked example. We concede that the richness of invariant algebras may differ dramatically between fundamental physics and social science, and that the thesis predicts this difference—mathematical models in physics are far more precise than in the social sciences, as the PFT requires.

10.8 The Dual-Source Decomposition Is Not a Theorem

Objection. The Dual-Source Decomposition (Proposition 9.10) is a philosophical classification dressed in formal notation. There is no algorithm for deciding which axioms of a physical theory belong to the cognitive component versus the empirical component.

Response. The objection has merit, and the result has been relabeled as a Proposition to reflect its status as a formalized philosophical argument rather than a purely mathematical theorem. The decomposition is not merely verbal, however: the formal component $\mathcal{T}_{\text{form}}$ has identifiable signatures (stability across theory change, shared across all physical theories, plausibly traceable to cognitive architecture), while the content component $\mathcal{T}_{\text{cont}}$ is specific to particular theories and determined by measurement. A fully rigorous criterion for distinguishing them—perhaps in terms of invariance under theory revision—remains an open problem.

10.9 Other Explanations Are Not Adequately Eliminated

Objection. The PFT has not been shown to be preferable to Tegmark’s MUH, structural realism, or evolutionary accounts. If the MUH is correct, it explains everything the PFT explains and more.

Response. We do not claim to have refuted any competing position. The PFT is compatible with several alternatives—if the MUH is true, the persistence filter is a selection principle within the mathematical multiverse. The advantage of the PFT is that it provides a mechanism (persistence) doing explanatory work regardless of one’s ontological commitments, and it generates testable predictions (Section 11) that give it empirical traction beyond purely philosophical accounts.

11 Falsifiable Predictions

A structural meta-thesis gains scientific credibility when it generates auxiliary predictions testable with available or foreseeable methods. We identify six predictions of the Persistence Filter Thesis, ordered by proximity to current experimental and computational capabilities.

11.1 Persistence–Tractability Correlation

Prediction 1. *Across a class of comparable systems, the degree of persistence (measured by stability margin, basin-of-attraction volume, or number of independent approximate invariants) should correlate positively with the quality of available mathematical descriptions.*

A concrete test: compare the mathematical tractability of long-lived atomic nuclei versus short-lived nuclear resonances. The PFT predicts that stable nuclei should admit more non-trivial conserved quantum numbers than highly unstable ones—and empirically, they do: stable nuclei respect more approximate symmetries (isospin, $SU(3)$ flavor) than resonance states, which decay too quickly for these approximate symmetries to be operationally meaningful. Making this quantitative—defining measures of “persistence” and “tractability” and demonstrating a systematic correlation—would constitute a direct test of the thesis.

11.2 Structural Change at Phase Transitions

Prediction 2. *When a system crosses a phase transition, the algebraic structure of its invariants should undergo a classifiable structural transformation—specifically, symmetry-breaking transitions should correspond to subalgebra inclusions.*

This prediction is partially captured by Landau theory [Landau, 1937], where the order parameter’s symmetry group reduces from G to a subgroup $H \subset G$ at the transition. The PFT predicts that this pattern holds more generally—including for non-equilibrium, topological, and dynamical transitions—and that the algebraic inclusion $\text{Inv}_G \hookrightarrow \text{Inv}_H$ (arising because G -invariants are automatically H -invariant) classifies the transition type. Systematic failures of this correspondence would weigh against the thesis.

11.3 Failure of Mathematical Modeling at Regime Transitions

Prediction 3. *Systems in genuine transition states—not near any attractor, not approximately conserving any quantities—should be the most resistant to effective mathematical modeling.*

This is perhaps the sharpest testable prediction. Candidates include economies in financial crisis, ecosystems during mass extinction events, and turbulent flows far from statistical steady state. If high-precision predictive mathematical models were routinely constructed for such systems, with no identifiable conserved quantities or invariant structures, this would constitute evidence against the thesis.

11.4 AI Discovery of Conservation Laws

Prediction 4. *Artificial learning systems trained to model physical dynamics from raw data should discover representations isomorphic to known conserved quantities (the $\mathcal{T}_{\text{cont}}$ component), even if they use a different formal scaffolding (the $\mathcal{T}_{\text{form}}$ component).*

This prediction follows from Remark 9.12 and the Dual-Source Decomposition. Recent work on symbolic regression [Udrescu and Tegmark, 2020] and Hamiltonian neural networks [Greydanus et al., 2019] provides partial evidence: these systems do discover energy conservation and other invariants from raw data. The prediction would be falsified if AI systems consistently found effective predictive models bearing no structural relation to known invariants.

11.5 Invariant Algebra Dimension Bounds Model Complexity

Prediction 5. *If a physical system has exactly k independent conserved quantities, then the minimal mathematical theory needed to describe its persistent behavior should have complexity scaling with k .*

This is a quantitative prediction: one should not need a theory of complexity vastly exceeding the dimensionality of the invariant algebra. If a system with two conserved quantities required an irreducibly infinite-dimensional theory for its description, this would challenge the thesis.

11.6 Fundamental Constants and Persistence Optimization

Prediction 6 (most speculative). *The values of fundamental physical constants should cluster near values that maximize the range and duration of persistent structures.*

This prediction overlaps with fine-tuning arguments but is distinct from anthropic reasoning: it does not say “constants are tuned for observers” but “constants are tuned for persistent structures of any kind.” This could in principle be tested by mapping out how the landscape of persistent structures—stable atoms, stable stars, stable bound states—varies as one varies fundamental constants, and checking whether observed values sit near maxima of persistence. While speculative, this prediction illustrates the thesis’s capacity to generate cosmological hypotheses.

12 Scope and Limitations

12.1 What the Argument Does and Does Not Show

The Persistence Filter Thesis establishes that every persistent physical system is necessarily a model of some formal mathematical theory. It does *not* establish:

- (a) That the specific mathematical theories used in current physics (quantum mechanics, general relativity, the Standard Model) are the *unique* theories compatible with the observed persistent structures. There may be alternative mathematical descriptions of the same invariant structure.
- (b) That the universe is “made of” mathematics in any ontological sense. The PFT is compatible with mathematical realism, nominalism, and structuralism alike.
- (c) That the particular values of physical constants (the fine-structure constant, the mass ratios of elementary particles, etc.) are explained by the persistence filter. The PFT explains why there is mathematical structure at all, and why certain *types* of structure recur, but the specific values of parameters within those structures remain contingent.
- (d) That topological or continuum structure is derived from persistence. Proposition 5.1 shows that persistence selects for continuous evolution *among systems on topological spaces*, but the topological structure of the state space itself—and the deeper question of why physical spacetime is a continuum—remain unexplained by the persistence filter alone. Similarly, Proposition 5.3 shows that persistent trajectories are variational critical points in *Lagrangian* systems, but whether a system admits a Lagrangian formulation is itself a structural feature not derived from persistence.

12.2 The Approximate Persistence Case

Our main results were stated for exactly persistent systems. In the physical world, most persistence is approximate: systems maintain their key properties within tolerances, not with infinite precision. The extension to approximate persistence requires replacing exact invariants with *approximate invariants* (sometimes called *adiabatic invariants* in physics):

Definition 12.1 (Approximate Invariant). An observable ϕ is an (ε, τ) -**approximate invariant** of (X, T, Φ) on A if:

$$\sup_{x \in A} \sup_{t \in [0, \tau]} \|\phi(\Phi(t, x)) - \phi(x)\| \leq \varepsilon. \quad (25)$$

The algebraic structure of approximate invariants is more delicate than that of exact invariants, and the extension of our results to this regime requires care. We develop the key properties.

Proposition 12.2 (Approximate Algebra of Approximate Invariants). *Let ϕ be an (ε_1, τ) -approximate invariant and ψ an (ε_2, τ) -approximate invariant of (X, T, Φ) on A . Then:*

- (a) $\phi + \psi$ is an $(\varepsilon_1 + \varepsilon_2, \tau)$ -approximate invariant.

(b) $c \cdot \phi$ is a $(|c|\varepsilon_1, \tau)$ -approximate invariant for any scalar $c \in \mathbb{R}$.

(c) $\phi \cdot \psi$ is an $(M_\psi \varepsilon_1 + M_\phi \varepsilon_2 + \varepsilon_1 \varepsilon_2, \tau)$ -approximate invariant, where $M_\phi = \sup_{x \in A} |\phi(x)|$ and $M_\psi = \sup_{x \in A} |\psi(x)|$.

Proof. Part (a) follows from the triangle inequality: $|(\phi + \psi)(\Phi(t, x)) - (\phi + \psi)(x)| \leq |\phi(\Phi(t, x)) - \phi(x)| + |\psi(\Phi(t, x)) - \psi(x)| \leq \varepsilon_1 + \varepsilon_2$.

Part (b) is immediate: $|c\phi(\Phi(t, x)) - c\phi(x)| = |c| \cdot |\phi(\Phi(t, x)) - \phi(x)| \leq |c|\varepsilon_1$.

For part (c), write $\phi' = \phi(\Phi(t, x))$ and $\psi' = \psi(\Phi(t, x))$, so $|\phi' - \phi| \leq \varepsilon_1$ and $|\psi' - \psi| \leq \varepsilon_2$. Then $|\phi'\psi' - \phi\psi| = |\phi'(\psi' - \psi) + \psi(\phi' - \phi)| \leq |\phi'|\varepsilon_2 + |\psi|\varepsilon_1 \leq (M_\phi + \varepsilon_1)\varepsilon_2 + M_\psi\varepsilon_1$. $\square$

This proposition shows that approximate invariants form an *approximate algebra*: the algebraic operations (addition, scalar multiplication, multiplication) are approximately closed, with error bounds that grow in a controlled way. The crucial feature is that the error bounds are *polynomial* in the individual tolerances—they do not blow up catastrophically.

Definition 12.3 (δ -Approximate Model). Let $\mathcal{T}$ be a first-order theory in a language with function symbols $f_1, \dots, f_n$ and relation symbols $R_1, \dots, R_m$. A δ -**approximate model** of $\mathcal{T}$ is a structure $\mathcal{M}_\delta$ in which every universal axiom $\forall \bar{x} \phi(\bar{x})$ of $\mathcal{T}$ is “approximately satisfied”: for every equation $s(\bar{x}) = t(\bar{x})$ that $\mathcal{T}$ entails, we have $\|s^{\mathcal{M}_\delta}(\bar{a}) - t^{\mathcal{M}_\delta}(\bar{a})\| \leq \delta$ for all tuples $\bar{a}$ from the domain.

Proposition 12.4 (Approximately Persistent Systems are Approximate Models). *Let (X, T, Φ) be an (ε, τ) -persistent system with approximate invariants $\phi_1, \dots, \phi_k$ and suppose the exact invariant algebra of an idealized version of the system satisfies polynomial identities $p_j(\phi_1, \dots, \phi_k) = 0$ for $j = 1, \dots, m$. Then the approximate system is a δ -approximate model of $\mathcal{T}_\mathfrak{A}$, where δ is bounded by a polynomial in ε whose degree depends on the degrees of the p_j .*

The significance of this extension is that it connects the precision of mathematical description to the precision of physical persistence. A system that is exactly persistent ($\varepsilon = 0$) is an exact model of $\mathcal{T}_\mathfrak{A}$. A system that is approximately persistent is a δ -approximate model, and the approximation quality δ degrades gracefully as ε increases. This explains a pattern familiar to working scientists: the precision of a mathematical model matches the precision of the invariances it exploits. Fundamental physics, where conservation laws hold to extraordinary precision, admits extraordinarily precise mathematical models. Biological and social systems, where invariants are approximate, admit correspondingly approximate models. The precision of mathematical description is not a free parameter—it is locked to the precision of persistence by the algebraic bounds of Proposition 12.2.

12.3 Open Problems

We identify several directions for further research:

1. **Quantitative bounds.** Can the persistence filter be made quantitative—that is, can we derive bounds on the complexity of the minimal mathematical theory required to describe a system with given persistence parameters (ε, τ) ?

2. **Emergence and hierarchy.** Persistent systems often exhibit hierarchical structure (atoms form molecules, molecules form cells, cells form organisms). How does the invariant algebra at one level of the hierarchy relate to the invariant algebra at another?
3. **Quantum foundations.** The argument in Section 5.4 for the necessity of Hilbert space structure is suggestive but not complete. Can the full axioms of quantum mechanics be derived from persistence conditions?
4. **Computability.** The persistence filter selects for mathematical describability, but does it select for *computable* describability? Is there a connection between persistence and Turing computability?
5. **Cosmological implications.** If the persistence filter is fundamental, does it have implications for the initial conditions of the universe or for the landscape of possible physical laws?

13 The Present Paper Within the Broader Research Programme

The Persistence Filter Thesis is not a standalone philosophical contribution. It is the foundational layer of a broader research programme, developed by the present author across more than eighty publications spanning 1999–2026, whose central claim is that persistence constitutes the primary organising principle of complex systems across all substrates: physical, biological, cognitive, and social [Kriger, 2026a]. The programme has been unified in a doctoral thesis [Kriger, 2026b] that synthesises these publications into a single deductive architecture. This section situates the present paper within that architecture and highlights results from the broader programme that bear directly on the arguments developed here.

13.1 Convergence Across Domains

The programme began with domain-specific investigations that independently converged on the same structural pattern. A metabolic model of the subjective acceleration of time perception [Kriger, 1999] identified persistence constraints on biological systems as the source of a measurable cognitive phenomenon. A quantitative framework for clinical decision structuring [Kriger et al., 2000] revealed that the reliability of clinical inference is bounded by the invariant structure of the decision space. A dynamical systems analysis of conflict escalation [Kriger, 2005] showed that the transition from stable institutional interaction to destructive conflict follows the formal pattern of a phase transition, with the persistence of cooperative equilibria governed by conserved institutional invariants.

In each case, the central finding was the same: the organisational logic of the system was determined not by its material substrate but by the formal constraints imposed by persistence. This convergence—across clinical medicine, cognitive science, and social theory—motivated the search for a unified theory.

13.2 Key Results

Several results from the broader programme connect directly to the present paper.

The Structural Genesis of Physical Law. Kriger [Kriger, 2018a] derives the form of Newton’s second law $F = ma$ —as a second-order deterministic evolution equation—not from empirical observation but from structural conditions on what it means for a system to persist in a metric space. The result reveals Newton’s laws as *theorems* about persistent metric systems rather than empirical postulates, exemplifying how the persistence filter operates at the level of specific physical laws.

The Four Laws of Self-Organising Systems. Kriger [Kriger, 2017] derives four formal laws governing any self-organising system that persists: bounds on self-organisation efficiency, minimal conditions for viability, observational asymmetry between functional and non-functional states, and interference laws for interacting persistent systems. Each law is derived from persistence conditions alone. The viability set of a self-organising system is precisely the persistent region A of Definition 3.2, and the four laws are structural consequences of the invariant algebra on A .

Cyclical Closure and Self-Sufficiency. Kriger [Kriger, 2024a,b] develops a topological analysis of self-sufficient systems using the first homology group H_1 of the system’s resource-flow graph, showing that self-sufficiency is equivalent to the existence of non-trivial one-cycles—closed loops carrying positive flux. Cyclical closure is itself a form of invariance: a closed resource cycle is a conserved circulating flux, and the algebraic structure of the space of such cycles is an instance of the invariant algebra of Proposition 3.5.

The Law of Imperative Uncertainty. Kriger [Kriger, 2025a] proves that any system capable of sustained complexity must permit persistent uncertainty and probabilistic deviation. The mathematically striking consequence is that the irreducible stochasticity of quantum mechanics and the sensitivity to initial conditions in chaotic systems are not failures of mathematical description but *requirements for persistence*. This enriches the PFT by showing that the filter selects not only for mathematical structure but for a specific kind—one that incorporates stochasticity as a constitutive element.

The Formalisation Laws. Kriger [Kriger, 2025b, 2026d] establishes three meta-mathematical laws governing formalisation itself: Total Formalisability, Boundary-Conditioned Verification, and Plurality and Incompleteness. Together with the present paper’s Persistence Filter Thesis, they yield the conclusion that mathematical describability of the physical world is necessary, but the specific description used is partly conventional and partly empirical—the formal content of the Dual-Source Decomposition of Section 9.3.

Astrophysical Applications. In a sequence of papers [Kriger, 2025c, 2026e], the persistence framework is applied to binary star formation, arguing that binary systems represent persistence-selected, energetically favoured configurations: the invariant quantities of a protostellar system (angular momentum, binding energy, magnetic flux) are more readily conserved in binary configurations. This demonstrates that the PFT generates specific, testable predictions in astrophysics.

Cognitive and Psychiatric Applications. The programme applies the persistence framework to perception, attention, and psychiatric phenomena [Kriger, 2026i,j], mapping DSM-5-TR diagnostic categories onto attractor topologies and bifurcation structures. In each case, the mathematical framework succeeds because the cognitive system is persistent,

its persistence is maintained by identifiable invariants, and those invariants carry algebraic structure isomorphic to the mathematical model.

13.3 Methodological Coherence

The broader programme is unified by a methodology the author terms the “Tree-Top Meta-Method” [Kriger, 2026]: formalise at the highest level of generality where results can still be rigorously derived, then descend to domain-specific applications. This methodology is itself a consequence of the persistence framework: substrate-independent formal constraints, precisely because they are invariant across domains, constitute the most persistent level of scientific knowledge.

The present paper occupies the apex of this hierarchy. It establishes the foundational result—the structural necessity of the isomorphism between persistent systems and formal theories—from which the domain-specific results descend as corollaries, each obtained by specifying the state space, the transformation group, and the observable family appropriate to the domain in question.

14 Conclusion

We have proposed and defended the Persistence Filter Thesis as a resolution to Wigner’s puzzle about the unreasonable effectiveness of mathematics in the natural sciences. The argument proceeds through a chain of necessary implications: persistence requires invariants; invariants carry algebraic structure; algebraic structures are models of formal theories. Therefore, the correspondence between mathematics and the physical world is not a contingent miracle but a structural necessity—a consequence of the simple fact that the physical systems we observe are the ones that persist.

This thesis unifies and extends several existing responses to Wigner’s puzzle. It incorporates the insights of structural realism (that structure is what matters), selection effects (that there is a filter at work), and mathematical Platonism (that the world has genuine mathematical structure)—while avoiding the ontological excesses of Platonism, the explanatory gaps of selection bias, and the limitations of evolutionary accounts.

The thesis also opens new avenues of investigation. By connecting mathematical effectiveness to persistence, it provides a framework for understanding why mathematics works differently in different domains (fundamental physics vs. biology vs. the social sciences), predicting where mathematical methods will be most successful, and identifying the structural features that make certain mathematical frameworks more “natural” than others.

The deepest implication may be this: the effectiveness of mathematics is unreasonable only if one begins with the assumption that mathematics and the physical world are fundamentally separate. The Persistence Filter Thesis suggests that they are not separate at all—not because the world is made of mathematics (as Tegmark proposes), but because mathematical structure is the inevitable signature of persistence, and persistence is the condition for existence.

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A Peer Reviews and Revision History

This appendix documents the peer review process and corresponding revisions made to this manuscript. It is maintained as a transparent record of scholarly dialogue and iterative improvement.

Version 1.0 — Initial Submission

- **Date:** February 2026
- **Status:** Submitted for peer review
- **Summary:** Original manuscript presenting the Persistence Filter Thesis as a resolution to Wigner’s puzzle, including the formal development of persistent systems theory, the Invariant Structure Theorem, the Persistence Filter Theorem, the functorial correspondence, and discussion of the specificity problem, comparison with existing approaches, interdisciplinary applications, and the Kantian counter-thesis with dual-source synthesis.

Version 2.0 — Revision After First Review

- **Date:** February 2026
- **Status:** Revised manuscript
- **Summary:** Major revision addressing all five major concerns and five minor concerns. Principal additions: Structural Richness Theorem (§4.2), Grounding of the Persistence Filter (§3.4), qualification of Noether’s theorem scope (§2.3), rigorous proofs throughout Section 5, recalibration of Section 8 results, expanded structural realism discussion (§6.2), corrected functor contravariance (§4.4), expanded approximate persistence theory (§9.2), and six new references.

First Round of Review

Reviewer Report

Overall assessment: Major revision recommended. The paper contains a worthwhile core argument, but requires revisions in several areas.

Major concerns raised:

1. *Vacuity of step (iii).* The observation that any structure is a model of some first-order theory is near-trivially true and does not require the completeness theorem. The real content must come from showing that persistent systems have *rich, non-trivial* algebraic structure. The gap between “model of some theory” and “described by deep, predictive mathematics” is precisely where Wigner’s puzzle lives.

2. *Circularity in the definition of persistence.* Persistence is defined in terms of invariant observables, but which observables count as relevant is left unformalized. This risks the circularity: persistent systems are those with invariants, invariants are what make systems persistent. The paper should address why physical systems have non-trivial conserved quantities in the first place.
3. *Overstatement of Noether’s theorem.* The paper claims symmetry and conservation are “mathematically equivalent,” but this holds only for continuous symmetries of Lagrangian systems. Since the paper claims generality extending to biology and economics, this mismatch must be addressed.
4. *Section 5 (specificity) underdeveloped.* Propositions 5.1, 5.3, 5.4 are largely informal. Proposition 5.1 assumes topological structure from the outset. Proposition 5.3 conflates dynamical stability with variational criticality. Section 5.4 is acknowledged as incomplete.
5. *Section 8 (Kantian counter-thesis) uneven.* The dual-source decomposition is philosophically appealing but the “theorem” is a conceptual framework in formal notation. Some results labeled “Theorem” are closer to formalized philosophical arguments. Theorem 8.6 relies on an imprecise probabilistic argument.

Minor concerns:

- a. Missing engagement with Pincock (2012), Bueno and French (2018), and the mapping account literature. The epistemic vs. ontic structural realism distinction is directly relevant but not discussed.
- b. The functor $\mathfrak{J}$ is contravariant (pullback reverses arrows); the paper should state this carefully and note that a functor does not establish “isomorphism” in the categorical sense without showing it is full, faithful, or part of an equivalence.
- c. Several results in Section 8 labeled “Theorem” would be more accurately labeled “Proposition.”
- d. The approximate persistence case deserves much more development.
- e. The peer review appendix with placeholder text should be populated.

Author Response and Revisions Made

1. **Vacuity of step (iii).** We have added Section 4.3, introducing the formal concept of *structural richness* (Definition 4.3) and proving the *Structural Richness Theorem* (Theorem 4.4), which shows that the invariant algebras of persistent systems are finitely axiomatizable (via the Hilbert Basis Theorem), have non-trivial consequences (constraining observables to algebraic varieties), and are multiply realizable. We have also added Proposition 4.6 on the depth gradient and revised the summary of the core argument in the Introduction to explicitly acknowledge the near-triviality of the bare model-theoretic observation and to direct the reader to the richness theorem.

2. **Circularity.** We have added Section 3.4, which provides three independent answers to the question of why physical systems possess invariants: a dynamical answer (symmetry as the default condition for spacetime, via Earman), a thermodynamic answer (conservation-law-compatible states dominate phase space, via Jaynes’ maximum entropy principle), and a logical answer (the persistence filter is analogous to natural selection—it explains the ubiquity of invariants without specifying which invariants, and this is not circular for the same reason that selection is not circular in biology).
3. **Noether overstatement.** We have substantially revised the discussion following the formal statement of Noether’s theorem to explicitly state that it applies only to continuous symmetries of Lagrangian systems, to note the distinct mechanisms for discrete symmetries (selection rules, quantum numbers) and for dissipative systems (attractors, Lyapunov functions), and to emphasize that the PFT does not depend on Noether’s theorem specifically but only on the existence of invariants by any mechanism.
4. **Section 5 strengthened.** Proposition 5.1 now has a complete proof (with explicit discussion of the topological presupposition). Proposition 5.3 now has a rigorous proof via second-order expansion of the action, with careful qualifications about its scope (Lagrangian systems only) and the distinction between critical points and stable critical points. Section 5.4 has been substantially expanded with two explicit propositions (Propositions 5.4 and 5.5) tracing the logical chain from persistence of transition probabilities through Wigner’s theorem and Stone’s theorem to Hilbert space structure, with honest acknowledgment of remaining gaps.
5. **Section 8 recalibrated.** Results in Section 8 that encode philosophical positions (Observational Indistinguishability, Predictive Excess, Cognitive Architecture Independence, Dual-Source Decomposition) have been relabeled as Propositions. An explicit disclaimer has been added at the first such result, noting that these are formalized philosophical arguments whose premises are not compelled by mathematics alone.
 - a. **Literature.** We have expanded Section 7.2 to discuss the epistemic vs. ontic structural realism distinction and to engage with the mapping account of Pincock (2012) and Bueno and French (2018). Six new references have been added.
 - b. **Functor contravariance.** Theorem 4.8 has been corrected to state the functor as $\mathcal{I} : \mathbf{Pers}^{\text{op}} \rightarrow \mathbf{Form}$ (contravariant), with the proof sketch updated and a new Remark 4.9 explicitly noting that the functor is not an equivalence of categories and explaining the sense in which “isomorphism” in the title should be understood.
 - c. Relabeling completed as noted in item 5 above.
 - d. **Approximate persistence.** Section 12.2 has been substantially expanded with Proposition 12.2 (approximate closure properties with explicit error bounds), Definition 12.3 (formal definition of δ -approximate models), and Proposition 12.4 (showing approximately persistent systems are approximate models with graceful degradation).
 - e. The peer review appendix has been populated with the present review and response.

Version 3.0 — Revision After Second Review

- **Date:** February 2026
- **Status:** Revised manuscript
- **Summary:** Major expansion addressing requests for toy models, falsifiable predictions, counterarguments, and literature engagement. Added: Section 6 (five illustrative models), Section 10 (nine counterarguments and responses), Section 11 (six falsifiable predictions), expanded Section 7 (Steiner’s tripartite distinction, conventionalism, information-theoretic approaches, constructor theory). Approximately 20 new references. Paper expanded from 40 to approximately 50 pages.

Version 4.0 — Revision After Third Review

- **Date:** February 2026
- **Status:** Revised manuscript
- **Summary:** Minor revision addressing three remaining major concerns and seven minor concerns from second-round review. Finite-generation assumption made explicit (Remark 4.5); grounding arguments clarified by domain (§3.4); depth gradient promoted to formal Proposition (Proposition 4.6); topological and Lagrangian limitations noted in Scope; natural selection disanalogy added. New Section on the broader research programme (Section 13) with 18 new references. Paper expanded to approximately 60 pages.

Second Round of Review

Reviewer Report

Overall assessment: Minor revision recommended. The paper has improved substantially and now makes a clear, well-qualified contribution.

Major concerns:

1. The Structural Richness Theorem assumes finite generation of the invariant algebra, which is not guaranteed by the persistence definition and may fail for biological and social systems.
2. The three answers to “why invariants?” (§3.4) operate at different levels and do not specify which answer applies to which class of systems.
3. The depth gradient (Remark 4.5) should be promoted from a remark to a formalized proposition using the approximate persistence machinery.

Minor concerns: (a) Topological presupposition in Prop. 5.1 not resolved; (b) Lagrangian assumption in Prop. 5.3 limits scope; (c) Hilbert space derivation starts from strong assumptions; (d) Natural selection analogy has a disanalogy worth noting; (e) Functor obstructions could be identified; (f) Technical clarification in Prop. 9.2(c); (g) Partial structures challenge not addressed.

Author Response and Revisions Made

1. **Finite generation.** The hypothesis is now stated explicitly in Theorem 4.4. A new Remark 4.5 discusses when finite generation holds (reductive groups, finite-dimensional Hamiltonian systems), when it may fail (infinite-dimensional, biological, social systems), and the available fallback (Gelfand spectrum, finite-dimensional approximations via approximate persistence theory).
 2. **Grounding coherence.** Section 3.4 now explicitly states which answer applies where: dynamical for fundamental Lagrangian physics, thermodynamic for macroscopic systems, logical for non-physical domains.
 3. **Depth gradient promoted.** The depth gradient is now Proposition 4.6, a formal Proposition with three quantitative claims linking algebraic richness (k, m, d) to model precision δ .
- a–b. **Topological and Lagrangian limitations** acknowledged in a new item (d) in Section 12.1.
- d. **Natural selection disanalogy** noted in Section 3.4: the filter constrains what is possible rather than selecting from what exists.

Additionally, a new Section 13 situates the paper within the broader research programme (80+ publications, 1999–2026), connecting the PFT to the structural genesis of physical law, the four laws of self-organising systems, cyclical closure, the law of imperative uncertainty, the formalisation laws, and applications to astrophysics and psychiatry.

Final Accepted Version

- **Date:** *[Placeholder]*
- **Summary of all changes from initial submission:** *[Placeholder]*